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Dirichlet's Theorem on the Infinitude of Primes

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Summer Project
May–July 2025

Contents

1	Dirichlet's Theorem for the Modulus 4: Preliminary Results	3
1.1	Dirichlet Characters	3
1.2	Dirichlet L -functions	3
1.3	Euler Product	4
1.4	Taking Logarithms	5
2	The Dirichlet Characters Modulo 4	6
2.1	The Principal Character	6
2.2	The Non-principal Character	6
2.3	The Associated L -functions	6
3	Logarithms of the Two L-functions	7
3.1	Expansion of $\log L(\chi_0, s)$	7
3.2	Expansion of $\log L(\chi_1, s)$	7
4	Adding the Two Logarithms	8
5	Convergence of the Higher-Order Terms	9
6	The Value of $L(\chi_1, 1)$	11
7	Behaviour of the Principal L-Function	11
8	Completion of the Proof for Primes Congruent to 1 Modulo 4	12
1	Characters of Finite Abelian Groups	16
1.1	Duality	16
1.2	Orthogonality Relation	23
2	Dirichlet Series	26
2.1	Ordinary Dirichlet Series: The Case $\lambda_n = \log n$	34
3	Zeta Functions and L-Functions	36
3.1	Euler Product	36
3.2	Euler Product	36
3.3	The L -functions	41
3.4	Product of the L -functions Relative to the Same Integer m	43
4	Density and Dirichlet Theorem	48

Abstract

The objective of this project is to prove Dirichlet's theorem on the infinitude of primes. Along the way, we will study Dirichlet L-functions and explore the orthogonality relations of characters, which play a crucial role in the proof.

Theorem 0.1 (Dirichlet). *Let a and m be relatively prime integers with $a \geq 1$ and $m \geq 1$. Then there exist infinitely many prime numbers p such that*

$$p \equiv a \pmod{m}.$$

1 Dirichlet's Theorem for the Modulus 4: Preliminary Results

Our goal is to prove that there are infinitely many primes congruent to 1 (mod 4). The proof proceeds by studying Dirichlet characters and their associated L -functions.

1.1 Dirichlet Characters

Let $N \geq 2$ be an integer.

Definition 1.1. A Dirichlet character modulo N is a function

$$\chi : \mathbb{Z} \longrightarrow \mathbb{C}$$

satisfying the following properties:

1. $\chi(n + N) = \chi(n)$ for every integer n .
2. $\chi(mn) = \chi(m)\chi(n)$ for all integers m, n .
3. If $(n, N) > 1$, then

$$\chi(n) = 0.$$

4. If $(n, N) = 1$, then

$$|\chi(n)| = 1.$$

Thus every Dirichlet character is periodic, completely multiplicative, and vanishes precisely on integers having a common factor with N .

1.2 Dirichlet L -functions

For a Dirichlet character χ , define

$$L(\chi, s) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}, \quad \Re(s) > 1.$$

Since

$$|\chi(n)| \leq 1,$$

we have

$$\sum_{n=1}^{\infty} \left| \frac{\chi(n)}{n^s} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^{\sigma}}, \quad \sigma = \Re(s) > 1.$$

The series on the right converges by the p -series test. Hence $L(\chi, s)$ converges absolutely for every $\Re(s) > 1$.

1.3 Euler Product

The complete multiplicativity of χ allows us to factor the Dirichlet series into an infinite product over primes.

Theorem 1.2 (Euler Product). *For every Dirichlet character,*

$$L(\chi, s) = \prod_p \left(1 - \frac{\chi(p)}{p^s} \right)^{-1}, \quad \Re(s) > 1.$$

Proof. Since

$$\left| \frac{\chi(p)}{p^s} \right| \leq \frac{1}{p^\sigma} < 1,$$

the geometric series gives

$$\left(1 - \frac{\chi(p)}{p^s} \right)^{-1} = 1 + \frac{\chi(p)}{p^s} + \frac{\chi(p)^2}{p^{2s}} + \dots.$$

Hence

$$\prod_p \left(1 - \frac{\chi(p)}{p^s} \right)^{-1} = \prod_p \left(1 + \frac{\chi(p)}{p^s} + \frac{\chi(p)^2}{p^{2s}} + \dots \right).$$

Because

$$\sum_p \frac{1}{p^\sigma} < \infty,$$

the above product converges absolutely.

Now expand the product.

Choosing

$$\frac{\chi(p)^{a_p}}{p^{a_p s}}$$

from the factor corresponding to the prime p gives the term

$$\prod_p \frac{\chi(p)^{a_p}}{p^{a_p s}}.$$

Only finitely many exponents a_p are non-zero.

By the Fundamental Theorem of Arithmetic,

$$n = \prod_p p^{a_p}$$

is the unique prime factorization of n .

Using complete multiplicativity,

$$\prod_p \chi(p)^{a_p} = \chi \left(\prod_p p^{a_p} \right) = \chi(n).$$

Similarly,

$$\prod_p p^{a_p s} = n^s.$$

Therefore every integer contributes exactly one term,

$$\frac{\chi(n)}{n^s},$$

and every such term appears exactly once.

Hence

$$\prod_p \left(1 - \frac{\chi(p)}{p^s}\right)^{-1} = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = L(\chi, s).$$

This proves the Euler product. □

1.4 Taking Logarithms

Since the Euler product converges absolutely when $\Re(s) > 1$, we may take logarithms term-by-term.

Thus,

$$\log L(\chi, s) = - \sum_p \log \left(1 - \frac{\chi(p)}{p^s}\right).$$

To expand the logarithm we use the power series

$$-\log(1 - x) = \sum_{n=1}^{\infty} \frac{x^n}{n}, \quad |x| < 1.$$

Since

$$\left| \frac{\chi(p)}{p^s} \right| \leq \frac{1}{p^\sigma} < 1,$$

we substitute

$$x = \frac{\chi(p)}{p^s}.$$

Therefore,

$$-\log \left(1 - \frac{\chi(p)}{p^s}\right) = \sum_{n=1}^{\infty} \frac{\chi(p)^n}{np^{ns}}.$$

Consequently,

$$\boxed{\log L(\chi, s) = \sum_p \sum_{n=1}^{\infty} \frac{\chi(p)^n}{np^{ns}}}.$$

This is the identity that will be used throughout the proof.

2 The Dirichlet Characters Modulo 4

There are exactly two Dirichlet characters modulo 4. We now describe them explicitly.

2.1 The Principal Character

The principal character modulo 4 is defined by

$$\chi_0(n) = \begin{cases} 1, & (n, 4) = 1, \\ 0, & (n, 4) > 1. \end{cases}$$

Since the integers relatively prime to 4 are precisely the odd integers, we have

$$\chi_0(n) = \begin{cases} 1, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

Hence

$$\chi_0(1) = 1, \quad \chi_0(2) = 0, \quad \chi_0(3) = 1, \quad \chi_0(4) = 0,$$

and the values repeat periodically with period 4.

2.2 The Non-principal Character

The unique non-principal Dirichlet character modulo 4 is

$$\chi_1(n) = \begin{cases} 1, & n \equiv 1 \pmod{4}, \\ -1, & n \equiv 3 \pmod{4}, \\ 0, & (n, 4) > 1. \end{cases}$$

Equivalently,

$$\chi_1(1) = 1, \quad \chi_1(2) = 0, \quad \chi_1(3) = -1, \quad \chi_1(4) = 0.$$

Again, these values repeat periodically.

2.3 The Associated L -functions

The Dirichlet L -function corresponding to the principal character is

$$L(\chi_0, s) = \sum_{n=1}^{\infty} \frac{\chi_0(n)}{n^s}.$$

Since $\chi_0(n) = 0$ whenever n is even, only odd integers contribute.

Hence

$$L(\chi_0, s) = 1 + \frac{1}{3^s} + \frac{1}{5^s} + \frac{1}{7^s} + \cdots.$$

Similarly,

$$L(\chi_1, s) = \sum_{n=1}^{\infty} \frac{\chi_1(n)}{n^s},$$

which becomes

$$L(\chi_1, s) = 1 - \frac{1}{3^s} + \frac{1}{5^s} - \frac{1}{7^s} + \frac{1}{9^s} - \cdots.$$

Notice that every odd integer congruent to 1 (mod 4) appears with a positive sign, while every odd integer congruent to 3 (mod 4) appears with a negative sign.

3 Logarithms of the Two L -functions

Using the logarithmic expansion proved previously,

$$\log L(\chi, s) = \sum_p \sum_{n=1}^{\infty} \frac{\chi(p)^n}{np^{ns}},$$

we compute the logarithms for the two characters separately.

3.1 Expansion of $\log L(\chi_0, s)$

Since

$$\chi_0(p) = \begin{cases} 1, & p \neq 2, \\ 0, & p = 2, \end{cases}$$

the prime 2 contributes nothing.

Therefore,

$$\log L(\chi_0, s) = \sum_{p \neq 2} \sum_{n=1}^{\infty} \frac{1}{np^{ns}}.$$

Separating the primes according to their residue classes modulo 4,

$$\log L(\chi_0, s) = \sum_{p \equiv 1(4)} \sum_{n=1}^{\infty} \frac{1}{np^{ns}} + \sum_{p \equiv 3(4)} \sum_{n=1}^{\infty} \frac{1}{np^{ns}}.$$

3.2 Expansion of $\log L(\chi_1, s)$

For the non-principal character,

$$\chi_1(2) = 0,$$

so again the prime 2 contributes nothing.

If

$$p \equiv 1 \pmod{4},$$

then

$$\chi_1(p) = 1.$$

Hence

$$\chi_1(p)^n = 1$$

for every positive integer n .
On the other hand, if

$$p \equiv 3 \pmod{4},$$

then

$$\chi_1(p) = -1,$$

and therefore

$$\chi_1(p)^n = (-1)^n.$$

Consequently,

$$\log L(\chi_1, s) = \sum_{p \equiv 1(4)} \sum_{n=1}^{\infty} \frac{1}{np^{ns}} + \sum_{p \equiv 3(4)} \sum_{n=1}^{\infty} \frac{(-1)^n}{np^{ns}}.$$

4 Adding the Two Logarithms

We now add the two identities obtained above.

For primes satisfying

$$p \equiv 1 \pmod{4},$$

both logarithms contribute

$$\sum_{n=1}^{\infty} \frac{1}{np^{ns}}.$$

Hence their sum equals

$$2 \sum_{n=1}^{\infty} \frac{1}{np^{ns}}.$$

Now suppose

$$p \equiv 3 \pmod{4}.$$

The corresponding contribution becomes

$$\sum_{n=1}^{\infty} \frac{1}{np^{ns}} + \sum_{n=1}^{\infty} \frac{(-1)^n}{np^{ns}} = \sum_{n=1}^{\infty} \frac{1 + (-1)^n}{np^{ns}}.$$

Since

$$1 + (-1)^n = \begin{cases} 2, & n \text{ even,} \\ 0, & n \text{ odd,} \end{cases}$$

only the even powers survive.

Therefore,

$$\sum_{n=1}^{\infty} \frac{1 + (-1)^n}{np^{ns}} = \sum_{m=1}^{\infty} \frac{1}{mp^{2ms}},$$

where we have written $n = 2m$.

Hence

$$\log L(\chi_0, s) + \log L(\chi_1, s) = 2 \sum_{p \equiv 1(4)} \sum_{n=1}^{\infty} \frac{1}{np^{ns}} + \sum_{p \equiv 3(4)} \sum_{m=1}^{\infty} \frac{1}{mp^{2ms}}.$$

Observe that the second double series contains only even powers of the primes $p \equiv 3 \pmod{4}$. We shall prove in the next section that this entire series converges absolutely even at $s = 1$. Thus the only possible source of divergence is the first double series, which involves the primes congruent to 1 $\pmod{4}$.

5 Convergence of the Higher-Order Terms

In the previous section we obtained

$$\begin{aligned} \log L(\chi_0, s) + \log L(\chi_1, s) &= 2 \sum_{p \equiv 1(4)} \sum_{n=1}^{\infty} \frac{1}{np^{ns}} \\ &\quad + \sum_{p \equiv 3(4)} \sum_{m=1}^{\infty} \frac{1}{mp^{2ms}}. \end{aligned}$$

We shall prove that the second double series converges even when $s = 1$.

Lemma 5.1. *The series*

$$\sum_p \sum_{n=2}^{\infty} \frac{1}{np^n}$$

converges.

Proof. Since $n \geq 2$,

$$\frac{1}{n} \leq 1,$$

and therefore

$$\sum_{n=2}^{\infty} \frac{1}{np^n} \leq \sum_{n=2}^{\infty} \frac{1}{p^n}.$$

The right-hand side is a geometric series:

$$\sum_{n=2}^{\infty} \frac{1}{p^n} = \frac{1/p^2}{1 - 1/p} = \frac{1}{p(p-1)}.$$

Hence

$$\sum_p \sum_{n=2}^{\infty} \frac{1}{np^n} \leq \sum_p \frac{1}{p(p-1)}.$$

Now observe that every prime satisfies $p \geq 2$, and

$$\frac{1}{p(p-1)} \leq \frac{1}{m(m-1)}$$

if we simply compare with all integers $m \geq 2$.

Thus

$$\sum_p \frac{1}{p(p-1)} \leq \sum_{m=2}^{\infty} \frac{1}{m(m-1)}.$$

Finally,

$$\frac{1}{m(m-1)} = \frac{1}{m-1} - \frac{1}{m}.$$

Therefore

$$\sum_{m=2}^N \frac{1}{m(m-1)} = 1 - \frac{1}{N}.$$

Letting $N \rightarrow \infty$,

$$\sum_{m=2}^{\infty} \frac{1}{m(m-1)} = 1.$$

Hence

$$\sum_p \sum_{n=2}^{\infty} \frac{1}{np^n} < \infty.$$

This proves the lemma. □

The previous lemma immediately implies that

$$\sum_{p \equiv 3(4)} \sum_{m=1}^{\infty} \frac{1}{mp^{2m}} < \infty.$$

Indeed,

$$\frac{1}{mp^{2m}} = \frac{1}{(2m)p^{2m}} + \frac{1}{(2m)p^{2m}} \leq 2 \frac{1}{(2m)p^{2m}},$$

and the right-hand side is bounded by the convergent double series of the preceding lemma.

Consequently,

$$\boxed{\sum_{p \equiv 3(4)} \sum_{m=1}^{\infty} \frac{1}{mp^{2m}} < \infty.}$$

Thus this term remains bounded as $s \rightarrow 1^+$.

6 The Value of $L(\chi_1, 1)$

We now evaluate the non-principal L -function at $s = 1$.

From its definition,

$$L(\chi_1, 1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots .$$

This is precisely the Gregory–Leibniz series.

By the classical expansion

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}, \quad |x| \leq 1,$$

we obtain, upon substituting $x = 1$,

$$\arctan(1) = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots .$$

Since

$$\arctan(1) = \frac{\pi}{4},$$

we conclude

$$\boxed{L(\chi_1, 1) = \frac{\pi}{4}.}$$

Hence

$$\boxed{\log L(\chi_1, 1) = \log \frac{\pi}{4},}$$

which is a finite real number.

7 Behaviour of the Principal L -Function

Recall that

$$L(\chi_0, s) = \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{1}{n^s}.$$

Since

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s},$$

we separate the even and odd integers.

The even part is

$$\sum_{n \text{ even}} \frac{1}{n^s} = \sum_{m=1}^{\infty} \frac{1}{(2m)^s} = \frac{1}{2^s} \zeta(s).$$

Therefore

$$L(\chi_0, s) = \zeta(s) - \frac{1}{2^s} \zeta(s) = \left(1 - \frac{1}{2^s}\right) \zeta(s).$$

Hence

$$\boxed{L(\chi_0, s) = \left(1 - 2^{-s}\right) \zeta(s).}$$

Taking logarithms,

$$\log L(\chi_0, s) = \log \zeta(s) + \log(1 - 2^{-s}).$$

As $s \rightarrow 1^+$,

$$\zeta(s) \longrightarrow \infty,$$

while

$$1 - 2^{-s} \longrightarrow \frac{1}{2}.$$

Thus

$$\log(1 - 2^{-s}) \longrightarrow \log \frac{1}{2},$$

which is finite.

Consequently,

$$\boxed{\log L(\chi_0, s) \longrightarrow \infty \quad (s \rightarrow 1^+).}$$

8 Completion of the Proof for Primes Congruent to 1 Modulo 4

Recall that

$$\log L(\chi, s) = \sum_p \sum_{n=1}^{\infty} \frac{\chi(p)^n}{np^{ns}}.$$

Separate the first term ($n = 1$) from the remaining terms:

$$\log L(\chi, s) = \sum_p \frac{\chi(p)}{p^s} + \sum_p \sum_{n=2}^{\infty} \frac{\chi(p)^n}{np^{ns}}.$$

Applying this to the principal character gives

$$\log L(\chi_0, s) = \sum_{p \neq 2} \frac{1}{p^s} + R_0(s),$$

where

$$R_0(s) = \sum_{p \neq 2} \sum_{n=2}^{\infty} \frac{1}{np^{ns}}.$$

By the lemma proved in the previous section,

$$R_0(1) < \infty.$$

Hence

$$R_0(s)$$

remains bounded as $s \rightarrow 1^+$.

Similarly,

$$\log L(\chi_1, s) = \sum_{p \equiv 1(4)} \frac{1}{p^s} - \sum_{p \equiv 3(4)} \frac{1}{p^s} + R_1(s),$$

where

$$R_1(s) = \sum_p \sum_{n=2}^{\infty} \frac{\chi_1(p)^n}{np^{ns}}.$$

Since

$$|\chi_1(p)^n| \leq 1,$$

we have

$$|R_1(s)| \leq \sum_p \sum_{n=2}^{\infty} \frac{1}{np^{ns}},$$

which converges at $s = 1$.

Therefore

$$R_1(s)$$

also remains bounded as $s \rightarrow 1^+$.

Now add the two logarithms.

Using

$$\chi_0(p) = 1 \quad (p \neq 2),$$

and

$$\chi_1(p) = \begin{cases} 1, & p \equiv 1 \pmod{4}, \\ -1, & p \equiv 3 \pmod{4}, \end{cases}$$

we obtain

$$\log L(\chi_0, s) + \log L(\chi_1, s) = \sum_{p \neq 2} \frac{1 + \chi_1(p)}{p^s} + R_0(s) + R_1(s).$$

Now observe that

$$1 + \chi_1(p) = \begin{cases} 2, & p \equiv 1 \pmod{4}, \\ 0, & p \equiv 3 \pmod{4}. \end{cases}$$

Hence

$$\log L(\chi_0, s) + \log L(\chi_1, s) = 2 \sum_{p \equiv 1(4)} \frac{1}{p^s} + R(s),$$

where

$$R(s) = R_0(s) + R_1(s)$$

is bounded near $s = 1$.

We now let

$$s \longrightarrow 1^+.$$

Since

$$\log L(\chi_0, s) \longrightarrow \infty,$$

and

$$\log L(\chi_1, s) = \log \left(\frac{\pi}{4} \right),$$

is finite, it follows that

$$\log L(\chi_0, s) + \log L(\chi_1, s) \longrightarrow \infty.$$

Therefore,

$$2 \sum_{p \equiv 1(4)} \frac{1}{p^s} + R(s) \longrightarrow \infty.$$

Since

$$R(s)$$

remains bounded,

$$\sum_{p \equiv 1(4)} \frac{1}{p^s} \longrightarrow \infty.$$

Finally, letting $s \rightarrow 1^+$,

$$\sum_{p \equiv 1(4)} \frac{1}{p} = \infty.$$

A convergent series cannot contain only finitely many non-zero terms. Indeed, if there were only finitely many primes satisfying

$$p \equiv 1 \pmod{4},$$

then

$$\sum_{p \equiv 1(4)} \frac{1}{p}$$

would be a finite sum, and hence finite.

This contradicts the divergence proved above.

Therefore there must be infinitely many primes congruent to 1 modulo 4.

$$\boxed{\#\{p \text{ prime} : p \equiv 1 \pmod{4}\} = \infty.}$$

The Theorem on Arithmetic Progressions

1 Characters of Finite Abelian Groups

1.1 Duality

Let G be a finite abelian group written multiplicatively.

Definition 1.1 (Character). *A character of a group G is a homomorphism from G into the multiplicative group $\mathbb{C}^*$ of nonzero complex numbers.*

Proposition 1.2. *The set of characters of a group G , i.e., $\text{Hom}(G, \mathbb{C}^*)$, forms an abelian group under pointwise multiplication.*

Proof. Define a binary operation

$$* : \text{Hom}(G, \mathbb{C}^*) \times \text{Hom}(G, \mathbb{C}^*) \longrightarrow \text{Hom}(G, \mathbb{C}^*)$$

by

$$(\chi_1, \chi_2) \mapsto \chi_1 * \chi_2,$$

such that

$$(\chi_1 * \chi_2)(t) = \chi_1(t) \cdot \chi_2(t), \quad \forall t \in G.$$

We want to verify that $\chi_1 * \chi_2 \in \text{Hom}(G, \mathbb{C}^*)$, i.e., that $\chi_1 * \chi_2$ is a homomorphism. Indeed, for all $t_1, t_2 \in G$,

$$\begin{aligned} (\chi_1 * \chi_2)(t_1 t_2) &= \chi_1(t_1 t_2) \cdot \chi_2(t_1 t_2) \\ &= (\chi_1(t_1) \cdot \chi_1(t_2)) \cdot (\chi_2(t_1) \cdot \chi_2(t_2)) \\ &= (\chi_1(t_1) \cdot \chi_2(t_1)) \cdot (\chi_1(t_2) \cdot \chi_2(t_2)) \\ &= (\chi_1 * \chi_2)(t_1) \cdot (\chi_1 * \chi_2)(t_2). \end{aligned}$$

Hence, $\chi_1 * \chi_2$ is a homomorphism, and the set $\text{Hom}(G, \mathbb{C}^*)$ is closed under pointwise multiplication. $\square$

We now verify that $\text{Hom}(G, \mathbb{C}^*)$ forms a group under this operation.

- **Associativity:** Pointwise multiplication is associative because multiplication in $\mathbb{C}^*$ is associative.
- **Existence of Identity:**
For every $\chi \in \text{Hom}(G, \mathbb{C}^*)$, there exists $\chi' \in \text{Hom}(G, \mathbb{C}^*)$ such that

$$\chi * \chi' = \chi.$$

Specifically, define $\chi_0(t) = 1$ for all $t \in G$. Then $\chi_0 \in \text{Hom}(G, \mathbb{C}^*)$, and for any χ ,

$$(\chi * \chi_0)(t) = \chi(t) \cdot \chi_0(t) = \chi(t) \cdot 1 = \chi(t).$$

So χ_0 is the identity character.

- **Existence of Inverse:**

For every $\chi \in \text{Hom}(G, \mathbb{C}^*)$, define $\chi'(t) = \frac{1}{\chi(t)}$. We want $\chi * \chi' = \chi_0$.

Then

$$(\chi * \chi')(t) = \chi(t) \cdot \chi'(t) = \chi(t) \cdot \frac{1}{\chi(t)} = 1 = \chi_0(t), \quad \forall t \in G.$$

We also check that $\chi' \in \text{Hom}(G, \mathbb{C}^*)$:

$$\chi'(t_1 t_2) = \frac{1}{\chi(t_1 t_2)} = \frac{1}{\chi(t_1) \chi(t_2)} = \frac{1}{\chi(t_1)} \cdot \frac{1}{\chi(t_2)} = \chi'(t_1) \cdot \chi'(t_2).$$

Hence χ' is a homomorphism, so the inverse exists in the group.

- **Commutativity:** Since $\mathbb{C}^*$ is abelian, the pointwise product is commutative:

$$(\chi_1 * \chi_2)(t) = \chi_1(t) \chi_2(t) = \chi_2(t) \chi_1(t) = (\chi_2 * \chi_1)(t).$$

Therefore, $\text{Hom}(G, \mathbb{C}^*)$ is an abelian group under pointwise multiplication.

Definition 1.3. The dual group of G , denoted by $\hat{G}$, is defined as

$$\hat{G} = \text{Hom}(G, \mathbb{C}^*),$$

where $\mathbb{C}^*$ denotes the multiplicative group of nonzero complex numbers.

Example 1.1. Let G be a cyclic group of order n with generator t , and let

$$\chi : G \rightarrow \mathbb{C}^*$$

be a character. Then, if $\omega = \chi(t)$, we have $\omega^n = 1$.

Proof. We compute:

$$\omega^n = (\chi(t))^n = \chi(t) \cdot \chi(t) \cdots \chi(t) = \chi(t^n) = \chi(1),$$

since $t^n = 1$ in G , and χ is a homomorphism. Therefore,

$$\omega^n = \chi(1) = 1.$$

Hence, $\chi(G) \subseteq \mathbb{C}^*$ is a finite cyclic subgroup, and in particular a subgroup of the group of n th roots of unity. $\square$

Example 1.2. Let G be a cyclic group of order n with generator t . Then, for every n th root of unity $\omega \in \mathbb{C}^*$, there exists a character $\chi : G \rightarrow \mathbb{C}^*$ such that

$$\chi(t^a) = \omega^a \quad \text{for all } a \in \mathbb{Z}.$$

Proof. The map defined by $\chi(t^a) = \omega^a$ is a homomorphism because

$$\chi(t^a \cdot t^b) = \chi(t^{a+b}) = \omega^{a+b} = \omega^a \cdot \omega^b = \chi(t^a) \cdot \chi(t^b).$$

Therefore, $\chi \in \text{Hom}(G, \mathbb{C}^*)$, i.e., χ is a character of G . $\square$

Example 1.3. Let $\widehat{G} = \text{Hom}(G, \mathbb{C}^*)$, where G is a cyclic group of order n with generator t . Then $\widehat{G}$ is isomorphic to μ_n , the group of n th roots of unity.

Proof. Define the map

$$\Phi : \widehat{G} = \text{Hom}(G, \mathbb{C}^*) \rightarrow \mu_n, \quad \Phi(\chi) = \chi(t),$$

where $G = \langle t \rangle$.

- **Well-defined:** If $\chi = \chi'$, then $\chi(t) = \chi'(t)$, so $\Phi(\chi) = \Phi(\chi')$.
- **Homomorphism:** For $\chi_1, \chi_2 \in \widehat{G}$,

$$\Phi(\chi_1 \cdot \chi_2) = (\chi_1 \cdot \chi_2)(t) = \chi_1(t) \cdot \chi_2(t) = \Phi(\chi_1) \cdot \Phi(\chi_2).$$

- **Injective:** Suppose $\Phi(\chi_1) = \Phi(\chi_2)$, i.e., $\chi_1(t) = \chi_2(t)$. Then, for all $a \in \mathbb{Z}$,

$$\chi_1(t^a) = (\chi_1(t))^a = (\chi_2(t))^a = \chi_2(t^a),$$

hence $\chi_1 = \chi_2$. So Φ is injective.

- **Surjective:** Given any $s \in \mu_n$, define $\chi : G \rightarrow \mathbb{C}^*$ by

$$\chi(t^a) = s^a.$$

This is a character of G (as shown in Example 2), and clearly $\Phi(\chi) = s$. Hence, Φ is surjective.

Therefore, Φ is an isomorphism, and $\widehat{G} \cong \mu_n$. □

Proposition 1.4. Let H be a subgroup of G . Then every character of H extends to a character of G .

Proof. We proceed by induction on the index $[G : H]$ of H in G .

Base case: If $[G : H] = 1$, then $G = H$, and the result is trivial.

Inductive step: Suppose $[G : H] > 1$. Then there exists $x \in G$ such that $x \notin H$.

Let $n > 1$ be the smallest integer such that $x^n \in H$. Let $\chi \in \text{Hom}(H, \mathbb{C}^*)$ be the given character on H .

Define $t := \chi(x^n)$. Since $\mathbb{C}^*$ is a divisible group, there exists $\omega \in \mathbb{C}^*$ such that $\omega^n = t$.

Let H' be the subgroup of G generated by H and x . Then every element $h' \in H'$ can be written uniquely as $h' = hx^a$, where $a \in \mathbb{Z}$ and $h \in H$.

Define a map $\chi' : H' \rightarrow \mathbb{C}^*$ by

$$\chi'(hx^a) := \chi(h)\omega^a.$$

We claim that χ' is a homomorphism. To verify this, let $h_1x^a, h_2x^b \in H'$. Then:

$$\begin{aligned} \chi'(h_1x^a \cdot h_2x^b) &= \chi'(h_1h_2x^{a+b}) \\ &= \chi(h_1h_2)\omega^{a+b} \\ &= \chi(h_1)\chi(h_2)\omega^a\omega^b \\ &= \chi'(h_1x^a)\chi'(h_2x^b). \end{aligned}$$

Thus, $\chi' \in \text{Hom}(H', \mathbb{C}^*)$. Since G is a finite abelian group, we can extend χ' to a character on all of G by using the inductive hypothesis. □

Remark 1.1. Define the map

$$\beta : \widehat{G} \rightarrow \widehat{H}, \quad \text{that is, } \beta : \text{Hom}(G, \mathbb{C}^*) \rightarrow \text{Hom}(H, \mathbb{C}^*)$$

by

$$\beta(\chi) = \chi|_H \quad \text{for } \chi \in \widehat{G} = \text{Hom}(G, \mathbb{C}^*).$$

Then

1. The restriction map β is a homomorphism.
2. β is surjective.
3. The kernel of β is the set of characters of G that are trivial on H .
4. $\ker \beta \cong \widehat{G/H}$, i.e., the kernel is isomorphic to the dual of G/H .

Proof. 1. Let $\chi_1, \chi_2 \in \widehat{G}$. Then,

$$\beta(\chi_1 \cdot \chi_2) = (\chi_1 \cdot \chi_2)|_H = \chi_1|_H \cdot \chi_2|_H = \beta(\chi_1) \cdot \beta(\chi_2),$$

so β is a homomorphism.

2. For surjectivity, this follows from the previous proposition, where we proved that every character on H can be extended to a character on G .

3. Now consider:

$$\ker \beta = \{ \chi \in \widehat{G} : \chi|_H = 1 \}.$$

Case I: If $\chi \in \ker \beta$, then for all $h \in H$, $\chi(h) = 1$. So χ is trivial on H .

Case II: Let χ be a character on G such that χ is trivial on H . Then clearly $\chi \in \ker \beta$.

Hence, $\ker \beta = \{ \chi \in \widehat{G} : \chi|_H = 1 \}$.

4. To prove that $\ker \beta \cong \widehat{G/H}$, define the map

$$\Phi : \widehat{G/H} = \text{Hom}(G/H, \mathbb{C}^*) \rightarrow \ker \beta$$

as follows: for $\phi \in \widehat{G/H}$, define $\Phi(\phi) = \chi$, where

$$\chi(g) = \phi(gH) \quad \text{for } g \in G.$$

We verify that Φ is a well-defined homomorphism.

Well-defined: Suppose $\phi = \phi'$, then for all $gH \in G/H$, $\phi(gH) = \phi'(gH) \Rightarrow \chi(g) = \chi'(g)$.

Injectivity: Suppose $\Phi(\phi_1) = \Phi(\phi_2)$. Then for all $g \in G$,

$$\phi_1(gH) = \chi_1(g) = \chi_2(g) = \phi_2(gH) \Rightarrow \phi_1 = \phi_2.$$

Homomorphism: Let $\phi_1, \phi_2 \in \widehat{G/H}$. Then for all $g \in G$,

$$\Phi(\phi_1 \cdot \phi_2)(g) = (\phi_1 \cdot \phi_2)(gH) = \phi_1(gH) \cdot \phi_2(gH) = \Phi(\phi_1)(g) \cdot \Phi(\phi_2)(g),$$

so $\Phi(\phi_1 \cdot \phi_2) = \Phi(\phi_1) \cdot \Phi(\phi_2)$.

Surjectivity: Let $\chi \in \ker \beta$. Then $\chi : G \rightarrow \mathbb{C}^*$ with $\chi|_H = 1$. So we can define $\phi \in \widehat{G/H}$ by

$$\phi(gH) = \chi(g),$$

which is well-defined and satisfies $\Phi(\phi) = \chi$.

Hence, $\widehat{G/H} \cong \ker \beta$. □

Proposition 1.5. *The group $\widehat{G}$ is a finite abelian group of the same order as G .*

Proof. From the previous remark, we have the following short exact sequence:

$$1 \rightarrow \ker \beta \rightarrow \widehat{G} \xrightarrow{\beta} \widehat{H} \rightarrow 1,$$

which is isomorphic to:

$$1 \rightarrow \widehat{G/H} \rightarrow \widehat{G} \rightarrow \widehat{H} \rightarrow 1.$$

We prove the proposition by induction on $|G|$.

Base Case: Suppose $|G| = 1$. Then $\widehat{G} \cong \{1\}$, so clearly $|\widehat{G}| = |G| = 1$.

Inductive Step: Let $|G| = n \geq 2$, and assume the result holds for all groups of order less than n .

If G is cyclic, then we have already seen that $\widehat{G} \cong \mu_n$, the group of n -th roots of unity, which also has order n . Thus, $|\widehat{G}| = |G|$.

Now suppose G is abelian but not cyclic. Choose a proper nontrivial cyclic subgroup $H \subsetneq G$. From the short exact sequence above, we have:

$$|\widehat{G}| = |\widehat{H}| \cdot |\widehat{G/H}|.$$

Since H is cyclic, $|\widehat{H}| = |H|$. Also, by the inductive hypothesis, since $|G/H| < n$, we have $|\widehat{G/H}| = |G/H|$.

Therefore,

$$|\widehat{G}| = |H| \cdot |G/H| = |G|,$$

as required. □

Proposition 1.6. *Let G_1 and G_2 be finite abelian groups such that $G = G_1 \times G_2$. Then*

$$\widehat{G} \cong \widehat{G_1} \times \widehat{G_2}.$$

Proof. Define a map

$$\Psi : \widehat{G} \longrightarrow \widehat{G_1} \times \widehat{G_2}$$

such that for each $\phi \in \widehat{G}$, we define

$$\Psi(\phi) = (\phi_1, \phi_2),$$

where

$$\phi_1 := \phi|_{G_1 \times \{e_2\}} \in \widehat{G_1}, \quad \phi_2 := \phi|_{\{e_1\} \times G_2} \in \widehat{G_2}.$$

We first show that Ψ is well-defined. Since $G = G_1 \times G_2$, we have

$$|G| = |G_1| \cdot |G_2| \quad \text{and hence} \quad |\widehat{G}| = |G| = |G_1| \cdot |G_2| = |\widehat{G_1}| \cdot |\widehat{G_2}|.$$

Therefore, it suffices to prove that Ψ is an injective homomorphism.

Injectivity: Suppose $\Psi(\phi) = \Psi(\phi')$. Then

$$(\phi_1, \phi_2) = (\phi'_1, \phi'_2) \Rightarrow \phi_1 = \phi'_1, \quad \phi_2 = \phi'_2.$$

This implies

$$\phi|_{G_1 \times \{e_2\}} = \phi'|_{G_1 \times \{e_2\}}, \quad \phi|_{\{e_1\} \times G_2} = \phi'|_{\{e_1\} \times G_2}.$$

Now, for any $(x, y) \in G_1 \times G_2$, using the fact that ϕ and ϕ' are homomorphisms, we have

$$\phi(x, y) = \phi((x, e_2)(e_1, y)) = \phi(x, e_2) \cdot \phi(e_1, y) = \phi_1(x) \cdot \phi_2(y),$$

and similarly,

$$\phi'(x, y) = \phi'_1(x) \cdot \phi'_2(y) = \phi_1(x) \cdot \phi_2(y).$$

Thus $\phi = \phi'$, and so Ψ is injective.

Homomorphism: Let $\phi_1, \phi_2 \in \widehat{G}$. Then,

$$\Psi(\phi_1 * \phi_2) = \left((\phi_1 * \phi_2)|_{G_1 \times \{e_2\}}, (\phi_1 * \phi_2)|_{\{e_1\} \times G_2} \right).$$

Now, for all $t \in G$, we have

$$(\phi_1 * \phi_2)(t) = \phi_1(t) \cdot \phi_2(t).$$

Therefore,

$$\Psi(\phi_1 * \phi_2) = \left(\phi_1|_{G_1 \times \{e_2\}} * \phi_2|_{G_1 \times \{e_2\}}, \phi_1|_{\{e_1\} \times G_2} * \phi_2|_{\{e_1\} \times G_2} \right).$$

Now for any $(t_1, t_2) \in G_1 \times G_2$, we compute:

$$\begin{aligned} & (\phi_1|_{G_1 \times \{e_2\}} * \phi_2|_{G_1 \times \{e_2\}}, \phi_1|_{\{e_1\} \times G_2} * \phi_2|_{\{e_1\} \times G_2})(t_1, t_2) \\ &= \left((\phi_1|_{G_1 \times \{e_2\}} \cdot \phi_2|_{G_1 \times \{e_2\}})(t_1), (\phi_1|_{\{e_1\} \times G_2} \cdot \phi_2|_{\{e_1\} \times G_2})(t_2) \right) \\ &= (\phi_1(t_1, e_2) \cdot \phi_2(t_1, e_2), \phi_1(e_1, t_2) \cdot \phi_2(e_1, t_2)) \\ &= \left((\phi_1|_{G_1 \times \{e_2\}}, \phi_1|_{\{e_1\} \times G_2})(t_1, t_2) \cdot (\phi_2|_{G_1 \times \{e_2\}}, \phi_2|_{\{e_1\} \times G_2})(t_1, t_2) \right). \end{aligned}$$

Hence,

$$\Psi(\phi_1 * \phi_2) = \Psi(\phi_1) * \Psi(\phi_2),$$

so Ψ is a group homomorphism.

Since Ψ is an injective homomorphism between finite abelian groups of the same order, it is an isomorphism. Therefore,

$$\widehat{G} \cong \widehat{G_1} \times \widehat{G_2}.$$

□

Proposition 1.7. *Let G be a finite abelian group. Then the dual group $\widehat{G}$ is (non-canonically) isomorphic to G .*

Proof. Given that G is a finite abelian group, by the Fundamental Theorem of Finite Abelian Groups (Structure Theorem), we have

$$G \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \mathbb{Z}_{n_3} \times \cdots \times \mathbb{Z}_{n_k},$$

where $\prod_{i=1}^k n_i = |G|$ and $n_i \mid n_{i+1}$ for $i = 1, \dots, k-1$.

Taking character groups (duals) on both sides, we obtain

$$\widehat{G} \cong \widehat{\mathbb{Z}_{n_1}} \times \widehat{\mathbb{Z}_{n_2}} \times \cdots \times \widehat{\mathbb{Z}_{n_k}}.$$

Each cyclic group $\mathbb{Z}_{n_i}$ is isomorphic to its character group $\widehat{\mathbb{Z}_{n_i}}$, so

$$\widehat{\mathbb{Z}_{n_i}} \cong \mathbb{Z}_{n_i} \quad \text{for all } i = 1, 2, \dots, k.$$

Therefore, we have

$$\widehat{G} \cong \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k} \cong G.$$

Hence, $\widehat{G} \cong G$. □

Proposition 1.8. *Let G be a finite abelian group. Then $\widehat{\widehat{G}} \cong G$ canonically.*

Proof. Define

$$\Psi : G \longrightarrow \widehat{\widehat{G}} := \text{Hom}(\widehat{G}, \mathbb{C}^\times)$$

by

$$\Psi(g)(\chi) := \chi(g) \quad \text{for all } \chi \in \widehat{G}.$$

To see that Ψ is a group homomorphism, observe that for any $g_1, g_2 \in G$, we have:

$$\Psi(g_1 g_2)(\chi) = \chi(g_1 g_2) = \chi(g_1) \chi(g_2) = \Psi(g_1)(\chi) \cdot \Psi(g_2)(\chi),$$

which implies

$$\Psi(g_1 g_2) = \Psi(g_1) \cdot \Psi(g_2).$$

Hence, Ψ is a group homomorphism.

Next, observe that

$$|\widehat{\widehat{G}}| = |\widehat{G}| = |G|.$$

So to prove Ψ is an isomorphism, it suffices to show that $\ker(\Psi) = \{e\}$, i.e., Ψ is injective.

Claim: $\ker(\Psi) = \{e\}$.

Suppose $x \in G$ such that $\Psi(x) = 1''$, the trivial character of $\widehat{\widehat{G}}$, which means

$$\Psi(x)(\chi) = \chi(x) = 1 \quad \text{for all } \chi \in \widehat{G}.$$

We must show that $x = e$.

Assume $x \neq e$. Let $H = \langle x \rangle \leq G$, a cyclic subgroup generated by x . Let $|H| = n$, so $x^n = e$.

We know how to define nontrivial characters on cyclic groups by sending $x \mapsto \zeta$, where ζ is an n th root of unity. Hence, there exists a nontrivial character $\chi' \in \widehat{H}$ such that $\chi'(x) \neq 1$.

By the earlier proposition (every character of a subgroup extends to a character of the group), χ' extends to a character $\chi \in \widehat{G}$ such that $\chi|_H = \chi'$. Therefore,

$$\chi(x) = \chi'(x) \neq 1,$$

contradicting the assumption that $\chi(x) = 1$ for all $\chi \in \widehat{G}$.

Hence, no such $x \neq e$ can exist, so $\ker(\Psi) = \{e\}$, and Ψ is injective.

Since Ψ is an injective homomorphism between two finite abelian groups of the same order, it is an isomorphism.

Therefore, $\widehat{\widehat{G}} \cong G$ canonically. □

1.2 Orthogonality Relation

Proposition 1.9. *Let $n = \text{Card}(G)$, and let $\chi \in \widehat{G}$. Then*

$$\sum_{x \in G} \chi(x) = \begin{cases} n & \text{if } \chi = 1 \text{ (the trivial character),} \\ 0 & \text{if } \chi \neq 1. \end{cases}$$

Proof. Let $\chi = 1$, the identity character. Then $\chi(x) = 1$ for all $x \in G$, so

$$\sum_{x \in G} \chi(x) = \sum_{x \in G} 1 = n.$$

Let $\chi \neq 1$. Then there exists $y \in G$ such that $\chi(y) \neq 1$. Observe:

$$\chi(y) \sum_{x \in G} \chi(x) = \sum_{x \in G} \chi(y)\chi(x) = \sum_{x \in G} \chi(yx) = \sum_{x \in G} \chi(x),$$

since $y \mapsto yx$ is a bijection of G .

So we have:

$$(\chi(y) - 1) \sum_{x \in G} \chi(x) = 0.$$

Since $\chi(y) \neq 1$, we conclude that

$$\sum_{x \in G} \chi(x) = 0.$$

□

Corollary 1.1. *Let $x \in G$. Then*

$$\sum_{\chi \in \widehat{G}} \chi(x) = \begin{cases} n & \text{if } x = 1, \\ 0 & \text{if } x \neq 1. \end{cases}$$

Proof. If $x = 1$, then for all $\chi \in \widehat{G}$, we have $\chi(x) = \chi(1) = 1$. Hence,

$$\sum_{\chi \in \widehat{G}} \chi(x) = \sum_{\chi \in \widehat{G}} 1 = n.$$

Let $x \neq 1$. Then there exists $\chi' \in \widehat{G}$ such that $\chi'(x) \neq 1$ (as $\widehat{G}$ separates points).

Now consider:

$$\chi'(x) \sum_{\chi \in \widehat{G}} \chi(x) = \sum_{\chi \in \widehat{G}} \chi'(x)\chi(x) = \sum_{\chi \in \widehat{G}} (\chi'\chi)(x) = \sum_{\psi \in \widehat{G}} \psi(x),$$

since $\chi \mapsto \chi'\chi$ is a bijection of $\widehat{G}$. Therefore,

$$(\chi'(x) - 1) \sum_{\chi \in \widehat{G}} \chi(x) = 0.$$

As $\chi'(x) \neq 1$, it follows that

$$\sum_{\chi \in \widehat{G}} \chi(x) = 0.$$

□

C. Modular Characters

- Let m be an integer ≥ 1 . Then, define

$$G(m) := (\mathbb{Z}/m\mathbb{Z})^\times,$$

the multiplicative group of invertible elements of the group $\mathbb{Z}/m\mathbb{Z}$.

- It is an abelian group of order $\varphi(m)$, where φ is Euler's phi function.
- An element $\chi \in \widehat{G(m)}$ is called a *character modulo m* .
- Since $\chi \in \widehat{G(m)}$ is defined only on integers coprime to m , we can extend such functions to all integers by defining

$$\chi(a) := 0 \quad \text{if } \gcd(a, m) \neq 1.$$

If n is an odd integer, define the elements $\varepsilon(n)$ and $\omega(n)$ in $\mathbb{Z}/2\mathbb{Z}$ as follows:

$$\varepsilon(n) = \frac{n-1}{2} \pmod{2} = \begin{cases} 0 & \text{if } n \equiv 1 \pmod{4}, \\ 1 & \text{if } n \equiv -1 \equiv 3 \pmod{4}, \end{cases}$$

$$\omega(n) = \frac{n^2-1}{8} \pmod{2} = \begin{cases} 0 & \text{if } n \equiv \pm 1 \pmod{8}, \\ 1 & \text{if } n \equiv \pm 5 \pmod{8}. \end{cases}$$

Example:

- (1) Let $m = 4$. Then

$$G(4) = (\mathbb{Z}/4\mathbb{Z})^\times = \{1, 3\} \cong \mathbb{Z}_2.$$

- We can think of characters as homomorphisms from $G(4)$ to the group of 2^{nd} roots of unity.
- There are exactly two such homomorphisms: one trivial, and one non-trivial.

The function

$$\varepsilon : (\mathbb{Z}/4\mathbb{Z})^\times \longrightarrow \mathbb{Z}/2\mathbb{Z}$$

is a group homomorphism.

Proof. We claim that for all $x, y \in (\mathbb{Z}/4\mathbb{Z})^\times$, we have

$$\varepsilon(xy) = \varepsilon(x) + \varepsilon(y).$$

Recall that

$$\varepsilon(z) = \frac{z-1}{2} \pmod{2}, \quad \text{for } z \in (\mathbb{Z}/4\mathbb{Z})^\times.$$

Then,

$$\frac{xy-1}{2} = \frac{(x-1)(y-1) + (x-1) + (y-1)}{2}.$$

Since $x, y \in (\mathbb{Z}/4\mathbb{Z})^\times$, they are invertible modulo 4, so x, y are odd integers.

Let $x - 1 = 2k_1$, and $y - 1 = 2k_2$, for some integers k_1, k_2 . Then,

$$(x - 1)(y - 1) = 4k_1k_2.$$

Substituting into the earlier expression, we get

$$\frac{xy - 1}{2} \equiv \frac{(x - 1)(y - 1)}{2} + \frac{x - 1}{2} + \frac{y - 1}{2} \pmod{2}.$$

Thus,

$$\varepsilon(xy) \equiv \varepsilon(x) + \varepsilon(y) \pmod{2},$$

which proves that ε is a group homomorphism. □

2. The function

$$\omega : (\mathbb{Z}/8\mathbb{Z})^\times \longrightarrow \mathbb{Z}/2\mathbb{Z}$$

is a group homomorphism.

Proof. We compute:

$$w(xy) = \frac{(xy)^2 - 1}{8} \equiv \frac{(x^2 - 1)(y^2 - 1)}{8} + \frac{x^2 - 1}{8} + \frac{y^2 - 1}{8} \pmod{2}$$

For all $x, y \in \{\pm 1, \pm 5\}$, we observe that:

$$\frac{(x^2 - 1)(y^2 - 1)}{8} \equiv 0 \pmod{2}$$

Therefore:

$$w(xy) \equiv \frac{x^2 - 1}{8} + \frac{y^2 - 1}{8} \pmod{2} \equiv w(x) + w(y) \pmod{2}$$

which establishes the additivity property. □

3. Non-trivial homomorphism (character)

$$\chi : G(4) \rightarrow \mathbb{C}^*$$

such that

$$\chi(x) = (-1)^{\varepsilon(x)},$$

where $\varepsilon(x)$ is a homomorphism from $G(4)$ to $\mathbb{Z}_2$.

4. Consider $m = 8$. Then

$$G(8) = \{1, 3, 5, 7\}.$$

We observe that

$$G(8) \cong \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The number of characters on $G(8)$ is given by

$$\gcd(2, 2) \cdot \gcd(2, 2) = 4.$$

So, there are 3 non-trivial characters. Let

$$\chi_i : G(8) \rightarrow \mathbb{C}^*, \quad i = 1, 2, 3,$$

be defined as follows:

1. $\chi_1(x) = (-1)^{\varepsilon(x)}$,
2. $\chi_2(x) = (-1)^{w(x)}$,
3. $\chi_3(x) = (-1)^{\varepsilon(x)+w(x)} = (\chi_1 \cdot \chi_2)(x)$,

where $\varepsilon(x)$ and $w(x)$ are suitable homomorphisms from $G(8)$ to $\mathbb{Z}_2$, corresponding to the two cyclic components of $G(8) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

These are the three non-trivial characters on $G(8)$.

2 Dirichlet Series

Theorem 2.1 (Cauchy–Goursat Theorem). *Let f be analytic in an open set $D \subset \mathbb{C}$, and let $z_1, z_2, z_3 \in D$. Suppose the closed triangle T with vertices z_1, z_2, z_3 is contained entirely within D . Then*

$$\int_{\partial T} f(z) dz = 0.$$

Theorem 2.2 (Morera’s Theorem). *Suppose f is continuous on an open set $D \subset \mathbb{C}$, and satisfies*

$$\int_C f(z) dz = 0$$

for every closed contour C contained in D . Then f is analytic on D .

Lemma 2.3. *Let $\Omega \subset \mathbb{C}$ be a domain, and let F be holomorphic in Ω . Given $\delta > 0$, define the set*

$$\Omega_\delta = \left\{ z \in \Omega : \overline{D_\delta(z)} \subset \Omega \right\}.$$

Then,

$$\sup_{z \in \Omega_\delta} |F'(z)| \leq \frac{1}{\delta} \sup_{t \in \Omega} |F(t)|.$$

Proof. Given $z \in \Omega_\delta$, the closed disc $\overline{D_\delta(z)} \subset \Omega$. Let $C_\delta(z)$ be the positively oriented boundary of this disc.

By the Cauchy Integral Formula,

$$F'(z) = \frac{1}{2\pi i} \int_{C_\delta(z)} \frac{F(t)}{(t-z)^2} dt.$$

Taking modulus,

$$|F'(z)| \leq \frac{1}{2\pi} \int_{C_\delta(z)} \frac{|F(t)|}{|t-z|^2} |dt|.$$

Since for all $t \in C_\delta(z)$, $|t-z| = \delta$ and the length of $C_\delta(z)$ is $2\pi\delta$, we have

$$|F'(z)| \leq \frac{1}{2\pi} \cdot \sup_{t \in \Omega} |F(t)| \cdot \int_{C_\delta(z)} \frac{1}{\delta^2} |dt| = \frac{1}{2\pi} \cdot \frac{1}{\delta^2} \cdot 2\pi\delta \cdot \sup_{t \in \Omega} |F(t)| = \frac{1}{\delta} \sup_{t \in \Omega} |F(t)|.$$

Hence,

$$\sup_{z \in \Omega_\delta} |F'(z)| \leq \frac{1}{\delta} \sup_{t \in \Omega} |F(t)|.$$

□

Lemma 2.4. *Let $D \subset \mathbb{C}$ be an open set, and let $\{f_n\}$ be a sequence of holomorphic functions on D which converges uniformly on every compact subset of D to a function f . Then f is holomorphic on D , and the sequence of derivatives $\{f'_n\}$ converges uniformly on compact subsets of D to f' , the derivative of f .*

Proof. We are given that:

- $f_n(z)$ are analytic functions on an open connected domain D ,
- $f_n(z) \rightarrow f(z)$ uniformly on every compact subset of D .

We claim that f is analytic on D .

Let $a \in D$. Since D is open, there exists $\delta > 0$ such that the closed ball $\overline{B(a, \delta)} \subset D$. Let T be a triangle contained in $B(a, \delta)$. Since $f_n \rightarrow f$ uniformly on compact subsets, we have

$$\int_{\partial T} f(z) dz = \lim_{n \rightarrow \infty} \int_{\partial T} f_n(z) dz.$$

But by the Cauchy–Goursat Theorem (stated below), we know that

$$\int_{\partial T} f_n(z) dz = 0 \quad \text{for all } n,$$

so

$$\int_{\partial T} f(z) dz = 0.$$

As f is continuous on D therefore, by Morera's theorem, f is analytic on D .

To show The sequence $f'_n(z)$ converges to $f'(z)$ uniformly on every compact subset of D will use the Lemma 2.3.

Let $F(t) = f_n(t) - f(t)$, and take $\Omega = D$. Then, using the lemma, we obtain:

$$\sup_{z \in D_\delta} |f'_n(z) - f'(z)| \leq \frac{1}{\delta} \sup_{t \in D} |f_n(t) - f(t)|.$$

Since by hypothesis $f_n \rightarrow f$ uniformly on compact subsets of D , we have

$$\sup_{t \in D} |f_n(t) - f(t)| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Hence,

$$\sup_{z \in D_\delta} |f'_n(z) - f'(z)| \rightarrow 0,$$

which shows that $f'_n \rightarrow f'$ uniformly on compact subsets of D . □

Lemma 2.5 (Abel's Lemma). *Let (a_n) and (b_n) be two sequences. Define*

$$A_{m,p} = \sum_{n=m}^p a_n, \quad \text{and} \quad S_{m,m'} = \sum_{n=m}^{m'} a_n b_n.$$

Then for all integers $m' > m$, we have

$$S_{m,m'} = \sum_{n=m}^{m'-1} A_{m,n} (b_n - b_{n+1}) + A_{m,m'} b_{m'}.$$

Proof (by Induction). Fix $m \in \mathbb{N}$, and define the proposition $P(m')$ for $m' > m$ as:

$$S_{m,m'} = \sum_{n=m}^{m'} a_n b_n = \sum_{n=m}^{m'-1} A_{m,n}(b_n - b_{n+1}) + A_{m,m'} b_{m'}.$$

Base Case: Let $m' = m + 1$. Then

$$S_{m,m+1} = a_m b_m + a_{m+1} b_{m+1},$$

$$A_{m,m} = a_m, \quad A_{m,m+1} = a_m + a_{m+1},$$

$$\sum_{n=m}^{m'} A_{m,n}(b_n - b_{n+1}) = A_{m,m}(b_m - b_{m+1}).$$

Thus,

$$A_{m,m}(b_m - b_{m+1}) + A_{m,m+1} b_{m+1} = a_m(b_m - b_{m+1}) + (a_m + a_{m+1})b_{m+1} = a_m b_m + a_{m+1} b_{m+1}.$$

So the base case holds.

Inductive Step: Assume $P(x)$ is true for some $x \geq m$, i.e.,

$$\sum_{k=m}^x a_k b_k = \sum_{n=m}^{x-1} A_{m,n}(b_n - b_{n+1}) + A_{m,x} b_x.$$

We want to show:

$$\sum_{k=m}^{x+1} a_k b_k = \sum_{n=m}^x A_{m,n}(b_n - b_{n+1}) + A_{m,x+1} b_{x+1}.$$

Proof: Let

$$A_{m,x+1} = \sum_{i=m}^{x+1} a_i.$$

Then:

$$\begin{aligned} \sum_{k=m}^{x+1} a_k b_k &= \sum_{k=m}^x a_k b_k + a_{x+1} b_{x+1} \\ &= \sum_{n=m}^{x-1} A_{m,n}(b_n - b_{n+1}) + A_{m,x} b_x + a_{x+1} b_{x+1} \\ &= \sum_{n=m}^{x-1} A_{m,n} b_n - \sum_{n=m}^{x-1} A_{m,n} b_{n+1} + A_{m,x} b_x + a_{x+1} b_{x+1} \\ &= \sum_{n=m}^x A_{m,n} b_n - \sum_{n=m}^x A_{m,n} b_{n+1} + A_{m,x} b_{x+1} + a_{x+1} b_{x+1} \\ &= \sum_{n=m}^x A_{m,n}(b_n - b_{n+1}) + (A_{m,x} + a_{x+1}) b_{x+1} \\ &= \sum_{n=m}^x A_{m,n}(b_n - b_{n+1}) + A_{m,x+1} b_{x+1}. \end{aligned}$$

Thus, the inductive step holds, and the lemma is proved. $\square$

Lemma 2.6. Let α, β be two real numbers such that $0 < \alpha < \beta$. Let $z = x + iy$ with $x, y \in \mathbb{R}$ and $x > 0$. Then

$$\left| e^{-\alpha z} - e^{-\beta z} \right| \leq \left| \frac{z}{x} \right| \left(e^{-\alpha x} - e^{-\beta x} \right).$$

Proof. Consider the integral

$$\int_{\alpha}^{\beta} e^{-tz} dt.$$

Let $z' = -tz$, so that $dt = \frac{dz'}{-z}$. Then:

$$\int_{\alpha}^{\beta} e^{-tz} dt = \int_{-\alpha z}^{-\beta z} e^{z'} \left(\frac{dz'}{-z} \right) = -\frac{1}{z} \int_{-\alpha z}^{-\beta z} e^{z'} dz' = \frac{1}{z} (e^{-\alpha z} - e^{-\beta z}).$$

Thus,

$$\int_{\alpha}^{\beta} e^{-tz} dt = \frac{1}{z} (e^{-\alpha z} - e^{-\beta z}),$$

or equivalently,

$$e^{-\alpha z} - e^{-\beta z} = z \int_{\alpha}^{\beta} e^{-tz} dt.$$

Taking modulus on both sides,

$$\left| e^{-\alpha z} - e^{-\beta z} \right| = |z| \left| \int_{\alpha}^{\beta} e^{-tz} dt \right| \leq |z| \int_{\alpha}^{\beta} |e^{-tz}| dt.$$

Now write $z = x + iy$. Then:

$$|e^{-tz}| = |e^{-t(x+iy)}| = |e^{-tx} e^{-ity}| = e^{-tx},$$

since $|e^{-ity}| = 1$. Therefore,

$$\left| \int_{\alpha}^{\beta} e^{-tz} dt \right| \leq \int_{\alpha}^{\beta} e^{-tx} dt = \left[\frac{e^{-tx}}{-x} \right]_{\alpha}^{\beta} = \frac{1}{x} (e^{-\alpha x} - e^{-\beta x}).$$

Hence,

$$\left| e^{-\alpha z} - e^{-\beta z} \right| \leq \frac{|z|}{x} (e^{-\alpha x} - e^{-\beta x}).$$

Since $x > 0$, we also note $|x| = x$, so:

$$\left| e^{-\alpha z} - e^{-\beta z} \right| \leq \left| \frac{z}{x} \right| (e^{-\alpha x} - e^{-\beta x}).$$

□

Dirichlet Series

Let (λ_n) be an increasing sequence of real numbers tending to $+\infty$.

For the sake of simplicity, we suppose that $\lambda_n \geq 0$.

A **Dirichlet series with exponents** (λ_n) is a series of the form

$$\sum a_n e^{-\lambda_n z}, \quad (a_n \in \mathbb{C}, z \in \mathbb{C}).$$

Example 2.1. Let $\lambda_n = \log n$. Then,

$$\sum a_n e^{-\lambda_n z} = \sum a_n e^{-(\log n)z} = \sum a_n n^{-z}.$$

This is the ordinary Dirichlet series.

Example 2.2. Let $\lambda_n = n$, and put $t = e^{-z}$. Then the series becomes

$$\sum a_n e^{-\lambda_n z} = \sum a_n t^n,$$

which is a power series in t .

Proposition 2.7. If the series $f(z) = \sum a_n e^{-\lambda_n z}$ converges for $z = z_0$, then it converges uniformly in every domain of the form

$$R(z - z_0) \geq 0, \quad \text{and} \quad \arg(z - z_0) \leq \alpha \quad \text{with } \alpha < \frac{\pi}{2}.$$

Here $R(z)$ denotes the real part of the complex number z .

Proof. Let $z = x + iy$, and suppose $z_0 \in \mathbb{C}$ is such that

$$f(z_0) = \sum a_n e^{-\lambda_n z_0}$$

converges.

Define a new Dirichlet series $q(z)$ by

$$q(z) := f(z + z_0),$$

so

$$q(z) = \sum a_n e^{-\lambda_n(z_0+z)} = \sum (a_n e^{-\lambda_n z_0}) e^{-\lambda_n z}.$$

Let $\alpha_n := a_n e^{-\lambda_n z_0}$. Then,

$$q(z) = \sum \alpha_n e^{-\lambda_n z}.$$

To prove the uniform convergence of $f(z)$, it suffices to show that $q(z)$ converges uniformly in regions of the form

$$\Omega_k := \left\{ z \in \mathbb{C} : R(z) \geq 0, \left| \frac{z}{R(z)} \right| \leq k \right\}, \quad \forall k > 0.$$

Let

$$A_{m,m'} = \sum_{n=m}^{m'} \alpha_n.$$

Since $\sum \alpha_n$ converges, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $m, m' \geq N$, we have

$$|A_{m,m'}| < \varepsilon. \tag{1}$$

Let $b_n := e^{-\lambda_n z}$ and consider the partial sum

$$S_{m,m'} := \sum_{n=m}^{m'} \alpha_n b_n = \sum_{n=m}^{m'} \alpha_n e^{-\lambda_n z}.$$

By using Abel's Lemma and the earlier exponential inequality, we have:

$$|S_{m,m'}| \leq \sum_{n=m}^{m'-1} |A_{m,n}| \cdot |e^{-\lambda_n z} - e^{-\lambda_{n+1} z}| + |A_{m,m'}| \cdot |e^{-\lambda_{m'} z}|. \quad (2)$$

Since $z = x + iy$ and $x = R(z) > 0$, and by using the exponential estimate

$$|e^{-\lambda_n z} - e^{-\lambda_{n+1} z}| \leq \left| \frac{z}{x} \right| (e^{-\lambda_n x} - e^{-\lambda_{n+1} x}),$$

we obtain

$$|S_{m,m'}| \leq \varepsilon \left(\left| \frac{z}{x} \right| \sum_{n=m}^{m'-1} (e^{-\lambda_n x} - e^{-\lambda_{n+1} x}) + 1 \right).$$

The telescoping sum yields:

$$\sum_{n=m}^{m'-1} (e^{-\lambda_n x} - e^{-\lambda_{n+1} x}) = e^{-\lambda_m x} - e^{-\lambda_{m'} x}. \quad (3)$$

Now, using the fact that $m' > m$ and λ_n is strictly increasing and positive, we have:

$$e^{-\lambda_{m'} x} < e^{-\lambda_m x} \Rightarrow e^{-\lambda_m x} - e^{-\lambda_{m'} x} < 1.$$

Hence,

$$|S_{m,m'}| < \varepsilon \left(\left| \frac{z}{x} \right| \cdot 1 + 1 \right) \leq \varepsilon(1 + k), \quad (4)$$

since $\left| \frac{z}{x} \right| \leq k$ in Ω_k .

Thus, $q(z) = \sum \alpha_n e^{-\lambda_n z}$ is uniformly Cauchy in Ω_k , and hence uniformly convergent.

Therefore,

$$f(z) = q(z - z_0) = \sum a_n e^{-\lambda_n z}$$

is uniformly convergent in every domain of the form

$$R(z - z_0) \geq 0, \quad \text{and} \quad \arg(z - z_0) \leq \alpha \quad \text{with} \quad \alpha < \frac{\pi}{2}.$$

□

Corollary 2.1. *If the Dirichlet series f converges at $z = z_0$, then it converges for all z such that $R(z) > R(z_0)$, and the function thus defined is holomorphic in that region.*

Proof. By the above proposition, if f converges at $z = z_0$, then it converges uniformly in every domain of the form

$$R(z - z_0) \geq 0, \quad \text{and} \quad |\arg(z - z_0)| \leq \alpha < \frac{\pi}{2}.$$

so f converges for all z such that $R(z) > R(z_0)$. Now, consider the partial sums of the Dirichlet series:

$$f_k(z) = \sum_{n=1}^k a_n e^{-\lambda_n z}.$$

Each f_k is holomorphic in the region $\text{Re}(z) > \text{Re}(z_0)$, and the sequence $\{f_k\}$ converges uniformly to f on compact subsets of this region.

By Lemma 2.4, since $\{f_k\}$ are holomorphic and converge uniformly to f on compact subsets of the region $\text{Re}(z) > \text{Re}(z_0)$, we conclude that f is also holomorphic in that region. □

Corollary 2.2. *The set of convergence of the series f contains a maximal open half-plane (called the half-plane of convergence).*

Corollary 2.3. *The function $f(z)$ converges to $f(z_0)$ as $z \rightarrow z_0$ in the domain*

$$R(z - z_0) \geq 0, \quad |\arg(z - z_0)| \leq \alpha \quad \text{with } \alpha < \frac{\pi}{2}.$$

Proof. By Proposition 2.7, if the series $\sum a_n e^{-\lambda_n z}$ converges at z_0 , then it converges uniformly in the domain

$$D = \left\{ z : \Re(z - z_0) \geq 0, \quad |\arg(z - z_0)| \leq \alpha < \frac{\pi}{2} \right\}.$$

Given that $z \rightarrow z_0$ in the domain D , and defining $q(z) = a_n e^{-\lambda_n z}$, which is a continuous function, we have:

$$q(z) \rightarrow q(z_0) = a_n e^{-\lambda_n z_0}.$$

That is,

$$\lim_{z \rightarrow z_0} a_n e^{-\lambda_n z} = a_n e^{-\lambda_n z_0}.$$

Since the series converges uniformly, we can exchange the limit and the summation. Therefore, as $z \rightarrow z_0$,

$$\lim_{z \rightarrow z_0} f(z) = \lim_{z \rightarrow z_0} \sum_{n=0}^{\infty} a_n e^{-\lambda_n z} = \sum_{n=0}^{\infty} \lim_{z \rightarrow z_0} a_n e^{-\lambda_n z} = \sum_{n=0}^{\infty} a_n e^{-\lambda_n z_0} = f(z_0).$$

Hence, $f(z) \rightarrow f(z_0)$ as $z \rightarrow z_0$ in the domain. □

Corollary 2.4. *The function $f(z) \equiv 0$ identically if and only if all its coefficients a_n are zero.*

Proof. Suppose

$$f(z) = \sum_{n=0}^{\infty} a_n e^{-\lambda_n z} = 0 \quad \text{for all } z \in \mathbb{C},$$

where (λ_n) is a strictly increasing sequence of non-negative real numbers.

We aim to prove that $a_n = 0$ for all $n \in \mathbb{N} \cup \{0\}$.

Step 1. Multiply both sides by $e^{\lambda_0 z}$:

$$e^{\lambda_0 z} f(z) = \sum_{n=0}^{\infty} a_n e^{-(\lambda_n - \lambda_0)z} = a_0 + \sum_{n=1}^{\infty} a_n e^{-(\lambda_n - \lambda_0)z}.$$

Then for any $z \in \mathbb{C}$, we have:

$$0 = e^{\lambda_0 z} f(z) = a_0 + \lim_{k \rightarrow \infty} \sum_{n=1}^k a_n e^{-(\lambda_n - \lambda_0)z}.$$

Letting $z \rightarrow \infty$ along the real axis and noting that $\lambda_n > \lambda_0$ for all $n \geq 1$, we obtain:

$$\lim_{z \rightarrow \infty} \left(\sum_{n=1}^k a_n e^{-(\lambda_n - \lambda_0)z} \right) = 0 \quad \text{for each fixed } k.$$

Therefore,

$$0 = \lim_{z \rightarrow \infty} \left(a_0 + \lim_{k \rightarrow \infty} \sum_{n=1}^k a_n e^{-(\lambda_n - \lambda_0)z} \right) = a_0.$$

Step 2. Multiply both sides by $e^{\lambda_1 z}$:

$$e^{\lambda_1 z} f(z) = \sum_{n=1}^{\infty} a_n e^{-(\lambda_n - \lambda_1)z} = a_1 + \sum_{n=2}^{\infty} a_n e^{-(\lambda_n - \lambda_1)z}.$$

Hence:

$$0 = e^{\lambda_1 z} f(z) = a_1 + \lim_{k \rightarrow \infty} \sum_{n=2}^k a_n e^{-(\lambda_n - \lambda_1)z}.$$

As before, letting $z \rightarrow \infty$ gives $a_1 = 0$.

Repeating this process: for any fixed $m \in \mathbb{N} \cup \{0\}$, multiply by $e^{\lambda_m z}$, then:

$$e^{\lambda_m z} f(z) = \sum_{n=m}^{\infty} a_n e^{-(\lambda_n - \lambda_m)z} = a_m + \sum_{n=m+1}^{\infty} a_n e^{-(\lambda_n - \lambda_m)z},$$

and write:

$$0 = a_m + \lim_{k \rightarrow \infty} \sum_{n=m+1}^k a_n e^{-(\lambda_n - \lambda_m)z}.$$

Letting $z \rightarrow \infty$, we conclude $a_m = 0$.

Hence, $a_n = 0$ for all n , and the result follows. $\square$

Dirichlet Series with Positive Coefficients

Proposition 2.8. *Let $f(z) = \sum a_n e^{-\lambda_n z}$ be a Dirichlet series whose coefficients a_n are real and $a_n \geq 0$. Suppose that f converges for $\Re(z) > \rho$, with $\rho \in \mathbb{R}$, and that the function f can be extended analytically to a function holomorphic in a neighborhood of the point $z = \rho$. Then there exists a number $\varepsilon > 0$ such that f converges for $\Re(z) > \rho - \varepsilon$.*

Proof. Given that

$$f(z) = \sum a_n e^{-\lambda_n z}$$

is a Dirichlet series such that:

- $a_n \geq 0$ and real for all n ,
- (λ_n) is a strictly increasing sequence of real numbers tending to ∞ ,
- f converges for $\Re(z) > \rho$,
- f admits an analytic continuation to a neighborhood of $z = \rho$.

Our goal is to show that there exists $\varepsilon > 0$ such that f converges for $\Re(z) > \rho - \varepsilon$.

We define a new function

$$g(z) := f(z + \rho) = \sum a_n e^{-\lambda_n(z+\rho)} = \sum a_n e^{-\lambda_n \rho} e^{-\lambda_n z}.$$

Define $\alpha_n := a_n e^{-\lambda_n \rho}$, so that

$$g(z) = \sum \alpha_n e^{-\lambda_n z}.$$

Then g is a Dirichlet series with nonnegative real coefficients, and it converges for $\Re(z) > 0$.

Since f has an analytic continuation to a neighborhood of $z = \rho$, g extends holomorphically to a neighborhood of $z = 0$. In particular, there exists $\varepsilon > 0$ such that g is holomorphic in the disc $|z - 1| \leq 1 + \varepsilon$.

Hence the Taylor series of $g(z)$ about $z = 1$ converges uniformly in $|z - 1| \leq 1 + \varepsilon$.

By Lemma 1 (previously established),

$$g^{(p)}(z) = \sum a_n (-\lambda_n)^p e^{-\lambda_n(z+\rho)} = \sum \alpha_n (-\lambda_n)^p e^{-\lambda_n z}, \quad \text{for } \Re(z) > 0.$$

In particular,

$$g^{(p)}(1) = \sum \alpha_n (-\lambda_n)^p e^{-\lambda_n}.$$

Therefore, the Taylor expansion of g around $z = 1$ is given by

$$g(z) = \sum_{p=0}^{\infty} \frac{1}{p!} (z-1)^p g^{(p)}(1), \quad |z-1| \leq 1 + \varepsilon.$$

In particular, for $z = -\varepsilon$, by using in series of positive term so we can interchange the summation we have

$$\begin{aligned} g(-\varepsilon) &= \sum_{p=0}^{\infty} \frac{(-1-\varepsilon)^p}{p!} g^{(p)}(1) \\ &= \sum_{p=0}^{\infty} \frac{(\varepsilon+1)^p}{p!} \sum \alpha_n \lambda_n^p e^{-\lambda_n} \\ &= \sum_n \alpha_n e^{-\lambda_n} \sum_{p=0}^{\infty} \frac{(\varepsilon+1)^p \lambda_n^p}{p!} \\ &= \sum_n \alpha_n e^{-\lambda_n} e^{(\varepsilon+1)\lambda_n} \\ &= \sum_n \alpha_n e^{\lambda_n \varepsilon} \\ &= \sum_n a_n e^{-\lambda_n \rho} e^{\lambda_n \varepsilon}. \end{aligned}$$

Thus,

$$g(-\varepsilon) = \sum a_n e^{-\lambda_n(\rho-\varepsilon)},$$

which is a convergent Dirichlet series. Therefore, the original series $f(z)$ converges at $z = \rho - \varepsilon$, and hence converges for $\Re(z) > \rho - \varepsilon$. $\square$

2.1 Ordinary Dirichlet Series: The Case $\lambda_n = \log n$

The corresponding Dirichlet series is of the form:

$$f(s) = \sum_{n=1}^{\infty} a_n e^{-\lambda_n s} = \sum_{n=1}^{\infty} a_n n^{-s} = \sum_{n=1}^{\infty} \frac{a_n}{n^s},$$

where s is the complex variable and $\lambda_n = \log n$. This is the standard form of an ordinary Dirichlet series.

Proposition 2.9. *If the sequence (a_n) is bounded, then the Dirichlet series $\sum_{n=1}^{\infty} \frac{a_n}{n^s}$ converges absolutely for $\Re(s) > 1$.*

Proof. Assume there exists a constant $M > 0$ such that $|a_n| \leq M$ for all n . Then,

$$\sum_{n=1}^{\infty} \left| \frac{a_n}{n^s} \right| = \sum_{n=1}^{\infty} \frac{|a_n|}{n^s} = \sum_{n=1}^{\infty} \frac{|a_n|}{n^{\Re(s)}} \leq M \sum_{n=1}^{\infty} \frac{1}{n^{\Re(s)}}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^{\Re(s)}}$ converges if and only if $\Re(s) > 1$. Therefore, the given Dirichlet series converges absolutely in the half-plane $\Re(s) > 1$. $\square$

Proposition 2.10. *If the partial sums $A_{m,p} = \sum_{n=m}^p a_n$ are bounded, then there is convergence (not necessarily absolute) for $\Re(s) > 0$.*

Proof. Consider

$$S_{m,m'} := \sum_{n=m}^{m'} a_n b_n, \quad \text{where } b_n = \frac{1}{n^s}.$$

Assume the partial sums $A_{m,p} = \sum_{n=m}^p a_n$ are bounded, i.e.,

$$|A_{m,p}| = \left| \sum_{n=m}^p a_n \right| \leq K.$$

Now by applying Abel's lemma, we have:

$$S_{m,m'} = \sum_{n=m}^{m'-1} A_{m,n} (b_n - b_{n+1}) + A_{m,m'} \cdot b_{m'}.$$

Substituting $b_n = \frac{1}{n^s}$, we get:

$$S_{m,m'} = \sum_{n=m}^{m'-1} \left(\sum_{k=m}^n a_k \right) \left(\frac{1}{n^s} - \frac{1}{(n+1)^s} \right) + A_{m,m'} \cdot \frac{1}{(m')^s}.$$

So,

$$|S_{m,m'}| \leq \sum_{n=m}^{m'-1} K \left| \frac{1}{n^s} - \frac{1}{(n+1)^s} \right| + K \cdot \left| \frac{1}{(m')^s} \right|.$$

Therefore,

$$|S_{m,m'}| \leq K \left(\sum_{n=m}^{m'-1} \left(\frac{1}{n^s} - \frac{1}{(n+1)^s} \right) + \frac{1}{(m')^s} \right).$$

Now, since $\Re(s) > 0$, we may assume $s \in \mathbb{R}$, by Proposition 2.7 If the convergence holds for real $s > 0$, then by earlier results it extends to all z with $\Re(z) > s$.

So take $s \in \mathbb{R}$ with $s > 0$. Then

$$n+1 > n \Rightarrow (n+1)^s > n^s > 1 \Rightarrow \frac{1}{(n+1)^s} < \frac{1}{n^s} < 1,$$

and hence

$$0 < \frac{1}{n^s} - \frac{1}{(n+1)^s} < 1, \quad \forall n, s > 0.$$

So,

$$\left| \frac{1}{n^s} - \frac{1}{(n+1)^s} \right| = \frac{1}{n^s} - \frac{1}{(n+1)^s},$$

$$\sum_{n=m}^{m'-1} \left(\frac{1}{n^s} - \frac{1}{(n+1)^s} \right) = \frac{1}{m^s} - \frac{1}{(m')^s}.$$

Putting this into the previous bound:

$$|S_{m,m'}| \leq K \left(\frac{1}{m^s} - \frac{1}{(m')^s} + \frac{1}{(m')^s} \right) = \frac{K}{m^s}.$$

So we obtain,

$$|S_{m,m'}| \leq \frac{K}{m^s}, \quad \text{where } s > 0.$$

Thus, for each fixed $s > 0$, we have

$$\frac{K}{m^s} \rightarrow 0 \quad \text{as } m \rightarrow \infty,$$

and hence

$$S_{m,m'} \rightarrow 0 \quad \text{as } m \rightarrow \infty \text{ and } m' \rightarrow \infty.$$

Therefore, if the partial sums $A_{m,p} = \sum_{n=m}^p a_n$ are bounded, then there is convergence for $\Re(s) > 0$. $\square$

3 Zeta Functions and L -Functions

3.1 Euler Product

Definition 3.1 (Multiplicative Function). *A function $f : \mathbb{N} \rightarrow \mathbb{C}$ is called multiplicative if it satisfies the following properties:*

- $f(1) = 1$,
- $f(mn) = f(m)f(n)$ for all integers $m, n \in \mathbb{N}$ such that $\gcd(m, n) = 1$.

Example 3.1. *The Euler phi function $\varphi(n)$ is an example of a multiplicative function.*

3.2 Euler Product

Assumption: Let f be a bounded multiplicative function.

Lemma 3.2. *The Dirichlet series*

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s}$$

converges absolutely for $\Re(s) > 1$, and its sum in this domain equals the convergent infinite product

$$\prod_{p \in \mathbb{P}} \left(\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} \right),$$

where $\mathbb{P}$ denotes the set of prime numbers.

Proof. By Proposition 2.9, since $f(n)$ is bounded and multiplicative, the Dirichlet series converges absolutely for $\Re(s) > 1$.

We now prove the Euler product identity.

Part (i): Finite Euler products.

For each integer $k \geq 1$, let $S_k := \{p_1, p_2, \dots, p_k\}$ denote the set of the first k prime numbers. Define

$$\mathcal{N}(S_k) := \{n \in \mathbb{N} : \text{every prime divisor of } n \text{ lies in } S_k\}.$$

Claim:

$$\sum_{n \in \mathcal{N}(S_k)} \frac{f(n)}{n^s} = \prod_{p \in S_k} \left(\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} \right).$$

Proof by induction on k :

Base case: Let $k = 1$, so $S_1 = \{p_1\}$. Then

$$\mathcal{N}(S_1) = \{p_1^0, p_1^1, p_1^2, \dots\},$$

and hence

$$\sum_{n \in \mathcal{N}(S_1)} \frac{f(n)}{n^s} = \sum_{m=0}^{\infty} \frac{f(p_1^m)}{p_1^{ms}},$$

which agrees with the product form.

Inductive step: Suppose the identity holds for $S_k = \{p_1, \dots, p_k\}$. Define $S_{k+1} := S_k \cup \{p_{k+1}\}$. Then

$$\mathcal{N}(S_{k+1}) = \{n \in \mathbb{N} : \text{all prime divisors of } n \text{ lie in } S_{k+1}\}.$$

Using multiplicativity of f , every $n \in \mathcal{N}(S_{k+1})$ can be uniquely written as $n = a \cdot p_{k+1}^m$, with $a \in \mathcal{N}(S_k)$ and $m \geq 0$. Thus,

$$\sum_{n \in \mathcal{N}(S_{k+1})} \frac{f(n)}{n^s} = \left(\sum_{n \in \mathcal{N}(S_k)} \frac{f(n)}{n^s} \right) \left(\sum_{m=0}^{\infty} \frac{f(p_{k+1}^m)}{p_{k+1}^{ms}} \right).$$

By the induction hypothesis,

$$\sum_{n \in \mathcal{N}(S_k)} \frac{f(n)}{n^s} = \prod_{p \in S_k} \left(\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} \right),$$

and so

$$\sum_{n \in \mathcal{N}(S_{k+1})} \frac{f(n)}{n^s} = \prod_{p \in S_{k+1}} \left(\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} \right).$$

Part (ii): Passing to the limit.

Since $\mathcal{N}(S_k) \nearrow \mathbb{N}$ as $k \rightarrow \infty$, we have

$$\lim_{k \rightarrow \infty} \sum_{n \in \mathcal{N}(S_k)} \frac{f(n)}{n^s} = \sum_{n=1}^{\infty} \frac{f(n)}{n^s}, \quad \text{for } \Re(s) > 1.$$

On the other hand, we also have

$$\lim_{k \rightarrow \infty} \prod_{p \in S_k} \left(\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} \right) = \prod_{p \in \mathbb{P}} \left(\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} \right),$$

since the infinite product on the right-hand side converges absolutely when f is bounded and $\Re(s) > 1$.

Conclusion:

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_{p \in \mathbb{P}} \left(\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} \right), \quad \text{for } \Re(s) > 1.$$

□

Lemma 3.3. *Let $f : \mathbb{N} \rightarrow \mathbb{C}$ be a strictly multiplicative function, i.e.,*

$$f(mn) = f(m)f(n) \quad \text{for all coprime } m, n \in \mathbb{N},$$

and suppose that the Dirichlet series

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s}$$

converges absolutely for $\Re(s) > 1$. Then in this region, we have the identity

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \frac{f(p)}{p^s}}.$$

Proof. Since f is strictly multiplicative, we have

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_{p \in \mathbb{P}} \left(\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} \right).$$

Now observe that since f is strictly multiplicative, it satisfies $f(p^m) = f(p)^m$ for all primes p and $m \geq 0$. Hence,

$$\sum_{m=0}^{\infty} \frac{f(p^m)}{p^{ms}} = \sum_{m=0}^{\infty} \left(\frac{f(p)}{p^s} \right)^m.$$

This is a geometric series that converges absolutely for $\Re(s) > 1$, since $|f(p)/p^s| < 1$. Thus,

$$\sum_{m=0}^{\infty} \left(\frac{f(p)}{p^s} \right)^m = \frac{1}{1 - \frac{f(p)}{p^s}}.$$

Substituting this into the product gives:

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^s} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \frac{f(p)}{p^s}}.$$

□

Zeta Function

If we have $f = 1$. We obtain the function

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \frac{1}{p^s}},$$

these formulas making sense for $\Re(s) > 1$.

Proposition 3.4. *The Riemann zeta function $\zeta(s)$ is holomorphic and nonzero in the half-plane $\Re(s) > 1$.*

One has

$$\zeta(s) = \frac{1}{s-1} + \phi(s),$$

where $\phi(s)$ is holomorphic for $\Re(s) > 0$.

Proof. by Proposition 2.9. The zeta function is absolutely convergent for $\Re(s) > 1$ so it is holomorphic and nonzero in the half-plane $\Re(s) > 1$

For other part We use the identity

$$\frac{1}{s-1} = \int_1^\infty \frac{1}{t^s} dt,$$

which holds for $\Re(s) > 1$.

Now, write

$$\begin{aligned} \zeta(s) &= \sum_{n=1}^\infty \frac{1}{n^s} = \frac{1}{s-1} + \sum_{n=1}^\infty \left(\frac{1}{n^s} - \int_n^{n+1} \frac{1}{t^s} dt \right) \\ &= \frac{1}{s-1} + \sum_{n=1}^\infty \phi_n(s), \end{aligned}$$

where we define

$$\phi_n(s) := \frac{1}{n^s} - \int_n^{n+1} \frac{1}{t^s} dt,$$

and hence

$$\phi(s) := \sum_{n=1}^\infty \phi_n(s).$$

Claim: The function $\phi(s)$ is well-defined and holomorphic for $\Re(s) > 0$.

Proof of claim: Each term $\phi_n(s)$ is holomorphic, and we show that the sum $\sum \phi_n(s)$ converges normally on compact subsets of $\{\Re(s) > 0\}$.

We estimate:

$$|\phi_n(s)| = \left| \int_n^{n+1} \left(\frac{1}{n^s} - \frac{1}{t^s} \right) dt \right| \leq \sup_{n \leq t \leq n+1} \left| \frac{1}{n^s} - \frac{1}{t^s} \right|.$$

Let $f(t) = \frac{1}{n^s} - \frac{1}{t^s}$, so $f'(t) = st^{-s-1}$. By the Mean Value Theorem, for some $c \in [n, n+1]$, and $t \in [n, n+1]$

$$\left| \frac{1}{n^s} - \frac{1}{t^s} \right| \leq \left| \frac{s}{c^{s+1}} \right| |n - t| \leq \frac{|s|}{n^{x+1}},$$

where $x = \Re(s)$.

Therefore,

$$|\phi_n(s)| \leq \frac{|s|}{n^{x+1}}.$$

So, the series

$$\sum_{n=1}^\infty |\phi_n(s)| \leq \sum_{n=1}^\infty \frac{|s|}{n^{x+1}}$$

converges for $x > 0$. Hence, the series $\sum \phi_n(s)$ converges uniformly on any compact subset of $\{\Re(s) > \varepsilon\}$ for any $\varepsilon > 0$. It follows that $\phi(s)$ is holomorphic for $\Re(s) > 0$.

Thus,

$$\zeta(s) = \frac{1}{s-1} + \phi(s), \quad \text{where } \phi(s) \text{ is holomorphic for } \Re(s) > 0.$$

□

Corollary 3.1. *The Riemann zeta function has a simple pole at $s = 1$.*

Corollary 3.2. *When $s \rightarrow 1^+$, one has*

$$\sum_p \frac{1}{p^s} \sim \log \frac{1}{s-1}, \quad \text{and} \quad \sum_{\substack{p \\ k \geq 2}} \frac{1}{p^{ks}} \text{ remains bounded.}$$

Proof. Recall that for $\Re(s) > 1$,

$$\zeta(s) = \prod_p \frac{1}{1 - \frac{1}{p^s}}.$$

Taking logarithms:

$$\log \zeta(s) = - \sum_p \log \left(1 - \frac{1}{p^s} \right) = \sum_p \left(\frac{1}{p^s} + \frac{1}{2p^{2s}} + \frac{1}{3p^{3s}} + \cdots \right).$$

Thus,

$$\log \zeta(s) = \sum_p \frac{1}{p^s} + \sum_{\substack{p \\ k \geq 2}} \frac{1}{kp^{ks}}.$$

Let us denote

$$\psi(s) := \sum_{\substack{p \\ k \geq 2}} \frac{1}{kp^{ks}}.$$

Now consider:

$$\sum_{\substack{p \\ k \geq 2}} \frac{1}{p^{ks}} = \sum_p \sum_{k \geq 2} \frac{1}{p^{ks}} = \sum_p \left(\sum_{k \geq 2} \frac{1}{p^{ks}} \right) = \sum_p \left(\frac{1}{p^{2s}} + \frac{1}{p^{3s}} + \cdots \right).$$

This is a geometric series:

$$\sum_{k \geq 2} \frac{1}{p^{ks}} = \frac{1}{p^{2s}} \cdot \frac{1}{1 - \frac{1}{p^s}} = \frac{1}{p^{2s}(1 - \frac{1}{p^s})} = \frac{1}{p^s(p^s - 1)}.$$

Hence,

$$\sum_{\substack{p \\ k \geq 2}} \frac{1}{p^{ks}} \leq \sum_p \frac{1}{p^s(p^s - 1)} \leq \sum_p \frac{1}{p(p-1)}.$$

Since the sum $\sum_{n=2}^{\infty} \frac{1}{n(n-1)} = 1$, we conclude that $\psi(s)$ remains bounded for $s > 1$.

Now from Proposition 3.4 (already proven), we know that

$$\zeta(s) = \frac{1}{s-1} + \phi(s),$$

where $\phi(s)$ is holomorphic for $\Re(s) > 0$. Then,

$$\frac{\zeta(s)}{\frac{1}{s-1}} = 1 + (s-1)\phi(s).$$

Hence,

$$\lim_{s \rightarrow 1^+} \frac{\zeta(s)}{\frac{1}{s-1}} = 1.$$

Taking logarithms on both sides,

$$\log \left(\frac{\zeta(s)}{\frac{1}{s-1}} \right) = \log \zeta(s) - \log \frac{1}{s-1},$$

so

$$\lim_{s \rightarrow 1^+} \left(\log \zeta(s) - \log \frac{1}{s-1} \right) = 0,$$

which implies

$$\lim_{s \rightarrow 1^+} \frac{\log \zeta(s)}{\log \frac{1}{s-1}} = 1.$$

Now, using the earlier identity

$$\log \zeta(s) = \sum_p \frac{1}{p^s} + \sum_{\substack{p \\ k \geq 2}} \frac{1}{k p^{ks}},$$

and dividing both sides by $\log \frac{1}{s-1}$, we get

$$\frac{\log \zeta(s)}{\log \frac{1}{s-1}} = \frac{\sum_p \frac{1}{p^s}}{\log \frac{1}{s-1}} + \frac{\psi(s)}{\log \frac{1}{s-1}}.$$

As $s \rightarrow 1^+$, the second term tends to 0 since $\psi(s)$ is bounded. Thus,

$$\lim_{s \rightarrow 1^+} \frac{\sum_p \frac{1}{p^s}}{\log \frac{1}{s-1}} = 1.$$

□

3.3 The L -functions

Let $m \geq 1$ be a fixed integer, and let χ be a Dirichlet character modulo m , i.e.,

$$\chi : G(m) = (\mathbb{Z}/m\mathbb{Z})^\times \longrightarrow \mathbb{C}^\times.$$

The corresponding L -function is defined by the Dirichlet series

$$L(s, \chi) := \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}.$$

The following properties of χ hold:

- If $\gcd(n, m) = 1$, then $\chi(n) \neq 0$.

- If $\gcd(n, m) > 1$, then $\chi(n) = 0$.
- χ is a multiplicative function, i.e., $\chi(mn) = \chi(m)\chi(n)$ whenever $\gcd(mn, m) = 1$.

Proposition 3.5. *For the trivial character $\chi = 1$, one has*

$$L(s, 1) = F(s) \cdot \zeta(s), \quad \text{where} \quad F(s) = \prod_{p|m} (1 - p^{-s}),$$

and in particular, $L(s, 1)$ extends analytically to the half-plane $\Re(s) > 0$, with a simple pole at $s = 1$.

Proof. by the Lemma 3.3, since Dirichlet characters χ are multiplicative, we have

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_{p \in \mathbb{P}} \left(1 - \frac{\chi(p)}{p^s} \right)^{-1}, \quad \text{for } \Re(s) > 1. \quad (1)$$

Now suppose $\chi = 1$, the trivial character modulo m . Then $\chi(p) = 0$ if $p \mid m$, and $\chi(p) = 1$ if $\gcd(p, m) = 1$. So the product (1) becomes:

$$L(s, 1) = \prod_{p|m} \left(1 - \frac{0}{p^s} \right)^{-1} \prod_{p \nmid m} \left(1 - \frac{1}{p^s} \right)^{-1} = \prod_{p|m} (1)^{-1} \prod_{p \nmid m} \left(1 - \frac{1}{p^s} \right)^{-1}.$$

Now rearranging,

$$L(s, 1) = \prod_{p|m} \left(1 - \frac{1}{p^s} \right) \cdot \prod_{p \in \mathbb{P}} \left(1 - \frac{1}{p^s} \right)^{-1} = F(s) \cdot \zeta(s),$$

where

$$F(s) := \prod_{p|m} \left(1 - \frac{1}{p^s} \right), \quad \text{and} \quad \zeta(s) = \prod_{p \in \mathbb{P}} \left(1 - \frac{1}{p^s} \right)^{-1}.$$

Now, by Proposition 3.4, we know that

$$\zeta(s) = \frac{1}{s-1} + \phi(s),$$

where $\phi(s)$ is holomorphic for $\Re(s) > 0$.

So,

$$L(s, 1) = F(s) \left(\frac{1}{s-1} + \phi(s) \right).$$

Since $F(s)$ is holomorphic for $\Re(s) > 0$, the product extends analytically to $\Re(s) > 0$, and $L(s, 1)$ has a simple pole at $s = 1$. $\square$

Proposition 3.6. *For $\chi \neq 1$, the Dirichlet series*

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$$

converges (respectively converges absolutely) in the half-plane $\Re(s) > 0$ (respectively $\Re(s) > 1$), and one has

$$L(s, \chi) = \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-s}}, \quad \text{for } \Re(s) > 1.$$

Proof. We first establish the convergence for $\Re(s) > 1$. Since χ is a bounded multiplicative function, Proposition 2.9 implies that $L(s, \chi)$ converges absolutely for $\Re(s) > 1$.

Moreover, as χ is also multiplicative in the strict sense, by Lemma 3.3 we obtain the Euler product:

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_{p \in \mathbb{P}} \frac{1}{1 - \chi(p)p^{-s}}, \quad \text{for } \Re(s) > 1.$$

It remains to show the convergence of $L(s, \chi)$ for $\Re(s) > 0$. For that it is enough to show the partial sums

$$A_{u,v} := \sum_{n=u}^v \chi(n), \quad \text{for } u \leq v.$$

is bounded

we consider

$$A_{u,v} := \sum_{n=u}^v \chi(n), \quad \text{for } u \leq v.$$

by Orthogonality of Characters i.e. Let $n = \text{Card}(G)$ and $\chi \in \widehat{G}$, then

$$\sum_{x \in G} \chi(x) = \begin{cases} n & \text{if } \chi = 1, \\ 0 & \text{if } \chi \neq 1. \end{cases}$$

Applying this to $\chi \neq 1$ and $\chi \in \widehat{G(m)}$, we get

$$\sum_{n=u}^{u+m-1} \chi(n) = 0,$$

Now,

$$A_{u,v} = \sum_{n=u}^v \chi(n) = \sum_{n=u}^{u+m-1} \chi(n) + \sum_{n=u+m}^{u+2m-1} \chi(n) + \cdots + \sum_{n=u+km}^v \chi(n),$$

where $v - (u + km) < m$.

Thus, the modulus satisfies:

$$|A_{u,v}| \leq \left| \sum_{n=u+km}^v \chi(n) \right| \leq m,$$

which is uniformly bounded.

Therefore, $A_{u,v}$ are bounded, and by the previous proposition, the series $L(s, \chi)$ converges for $\Re(s) > 0$. $\square$

3.4 Product of the L -functions Relative to the Same Integer m

- Let $m \geq 1$ be a fixed integer.
- Let $\bar{p} \in G(m)$ denote the class of a prime p modulo m , where $p \nmid m$. That is, $\bar{p}$ is the image of p in the group

$$G(m) := (\mathbb{Z}/m\mathbb{Z})^\times.$$

- Let $f = f(p)$ denote the order of $\bar{p}$ in the group $G(m)$.
 - By definition, $f(p)$ is the smallest positive integer $f > 1$ such that

$$p^f \equiv 1 \pmod{m}.$$

- Let $H := \langle \bar{p} \rangle$ be the cyclic subgroup of $G(m)$ generated by $\bar{p}$.
- Then the quotient group $G(m)/\langle \bar{p} \rangle$ is well-defined, and its order is

$$g(p) := \frac{\varphi(m)}{f(p)},$$

where $\varphi(m) = \#G(m)$ is Euler's phi function.

Lemma 3.7. *Let $p \nmid m$. Then we have the identity:*

$$\prod_{\chi \in \widehat{G(m)}} (1 - \chi(\bar{p})T) = \left(1 - T^{f(p)}\right)^{g(p)},$$

where the product ranges over all Dirichlet characters χ modulo m , and T is a complex variable.

Proof. Let $H = \langle \bar{p} \rangle \subseteq G(m)$. Then we have the natural restriction map:

$$\beta : \widehat{G(m)} \rightarrow \widehat{H}, \quad \beta(\chi) = \chi|_H.$$

This is a group homomorphism (since restriction respects group operation), and it is surjective.

Also, we know:

$$|\widehat{G(m)}| = |G(m)| = \varphi(m).$$

Define:

$$\mathcal{W} = \{\omega \in \mathbb{C}^\times : \omega^f = 1\},$$

i.e., the set of f -th roots of unity.

Define the map:

$$\Psi : \widehat{G(m)} \rightarrow \mathcal{W}, \quad \Psi(\chi) = \chi(\bar{p}).$$

Since $H = \langle \bar{p} \rangle$ has order f , we have $\widehat{H} \cong \mathcal{W}$

As every character of H extends to $G(m)$ (by surjectivity of β), the map:

$$\Psi : \widehat{G(m)} \rightarrow \mathcal{W}, \quad \Psi(\chi) = \chi(\bar{p}),$$

is a surjective group homomorphism.

Moreover:

$$\Psi(\chi \cdot \chi') = \chi(\bar{p}) \cdot \chi'(\bar{p}) = \Psi(\chi)\Psi(\chi').$$

Thus,

$$\ker(\Psi) = \{\chi \in \widehat{G(m)} : \chi(\bar{p}) = 1\}.$$

By the First Isomorphism Theorem, we get:

$$\widehat{G(m)} / \ker(\Psi) \cong \mathcal{W}.$$

Hence:

$$|\ker(\Psi)| = \frac{|\widehat{G(m)}|}{|\mathcal{W}|} = \frac{\varphi(m)}{f} = g(p).$$

Since each coset has the same size, for each $\omega \in \mathcal{W}$, there are exactly $g(p)$ characters $\chi \in \widehat{G(m)}$ such that $\chi(\bar{p}) = \omega$.

Therefore:

$$\prod_{\chi \in \widehat{G(m)}} (1 - \chi(\bar{p})T) = \prod_{\omega \in \mathcal{W}} (1 - \omega T)^{g(p)}.$$

Now using the identity from algebra that:

$$\prod_{\omega \in \mathcal{W}} (1 - \omega T) = 1 - T^f,$$

we obtain:

$$\prod_{\omega \in \mathcal{W}} (1 - \omega T)^{g(p)} = (1 - T^f)^{g(p)} = (1 - T^{f(p)})^{g(p)}.$$

This completes the proof. □

Proposition 3.8. *Define a new function $\zeta_m(s)$ by the formula*

$$\zeta_m(s) = \prod_{\chi \in \widehat{G(m)}} L(s, \chi).$$

Then one has

$$\zeta_m(s) = \prod_{p \nmid m} \left(\frac{1}{1 - \frac{1}{p^{f(p)s}}} \right)^{g(p)},$$

where this defines a Dirichlet series with positive integer coefficients converging in the half-plane $\Re(s) > 1$.

Proof. From Proposition 3.6, we have that for $\Re(s) > 1$,

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_{p \nmid m} \frac{1}{1 - \frac{\chi(p)}{p^s}}.$$

Therefore,

$$\zeta_m(s) = \prod_{\chi \in \widehat{G(m)}} L(s, \chi) = \prod_{\chi \in \widehat{G(m)}} \left(\prod_{p \nmid m} \frac{1}{1 - \frac{\chi(p)}{p^s}} \right).$$

We can rearrange the order of products:

$$= \prod_{p \nmid m} \left(\prod_{\chi \in \widehat{G(m)}} \frac{1}{1 - \frac{\chi(p)}{p^s}} \right).$$

Now consider. Let $T = p^{-s}$, and apply Lemma 3.7:

$$\prod_{\chi \in \widehat{G(m)}} (1 - \chi(p)T) = (1 - T^{f(p)})^{g(p)},$$

where $f(p)$ is the order of $\bar{p} \in G(m)$, and $g(p) = \frac{\varphi(m)}{f(p)}$.

Taking reciprocals, we obtain:

$$\prod_{\chi \in \widehat{G(m)}} \frac{1}{(1 - \chi(p)p^{-s})} = \frac{1}{(1 - p^{-sf(p)})^{g(p)}}.$$

Therefore,

$$\zeta_m(s) = \prod_{p \nmid m} \frac{1}{(1 - p^{-sf(p)})^{g(p)}}.$$

Now, observe that

$$\left(1 - \frac{1}{p^{f(p)s}}\right)^{-g(p)}$$

which is a Dirichlet series with positive coefficients.

Therefore, $\zeta_m(s)$ is a Dirichlet series with positive integral coefficients, and converges absolutely in the half-plane $\Re(s) > 1$, completing the proof. $\square$

Theorem 3.9.

1. ζ_m has a simple pole at $s = 1$.
2. $L(1, \chi) \neq 0$ for all $\chi \neq 1$.

Proof. We first prove that (b) $\Rightarrow$ (a).

We have the factorization

$$\zeta_m(s) = \prod_{\chi \in \widehat{G(m)}} L(s, \chi) = L(s, 1) \prod_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} L(s, \chi).$$

By Proposition 3.5, $L(s, 1)$ has a simple pole at $s = 1$. By Proposition 3.6, for $\chi \neq 1$, the Dirichlet series $L(s, \chi)$ converges in the half-plane $\Re(s) > 0$, and hence defines a holomorphic function there. In particular, the product

$$\prod_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} L(s, \chi)$$

is holomorphic and nonzero in a neighborhood of $s = 1$. It follows that $\zeta_m(s)$ has a simple pole at $s = 1$.

We now prove (a) $\Rightarrow$ (b) by contrapositive.

Suppose $\zeta_m(s)$ has a simple pole at $s = 1$ and assume, for contradiction, that there exists $\chi \neq 1$ such that $L(1, \chi) = 0$. As before, we have:

$$\zeta_m(s) = L(s, 1) \prod_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} L(s, \chi).$$

By Proposition 3.5, we can write:

$$\zeta_m(s) = F(s) \cdot \zeta(s) \cdot \prod_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} L(s, \chi),$$

where $\zeta(s) = \frac{1}{s-1} + \phi(s)$ and $F(s) = \prod_{p|m} (1 - p^{-s})$, as in Proposition 3.4. Here, $\phi(s)$ is holomorphic in a neighborhood of $s = 1$.

Since $\zeta_m(s)$ has a simple pole at $s = 1$, we must have:

$$\lim_{s \rightarrow 1} (s-1) \zeta_m(s)$$

exists and is nonzero. Using the expression above, we compute:

$$\begin{aligned} \lim_{s \rightarrow 1} (s-1) \zeta_m(s) &= \lim_{s \rightarrow 1} (s-1) F(s) \left(\frac{1}{s-1} + \phi(s) \right) \prod_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} L(s, \chi). \\ &= \lim_{s \rightarrow 1} F(s) (1 + (s-1)\phi(s)) \prod_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} L(s, \chi). \end{aligned}$$

But under our assumption, there exists some $\chi \neq 1$ such that $L(1, \chi) = 0$, and hence the product vanishes at $s = 1$. Therefore,

$$\lim_{s \rightarrow 1} (s-1) \zeta_m(s) = 0,$$

contradicting the fact that $\zeta_m(s)$ has a simple pole at $s = 1$.

Hence, $L(1, \chi) \neq 0$ for all $\chi \neq 1$. □

Proof of (b) in Theorem 1. Suppose, for the sake of contradiction, that there exists a nontrivial character $\chi \neq 1$ such that $L(1, \chi) = 0$. Then from the expression

$$\zeta_m(s) = \prod_{\chi \bmod m} L(s, \chi),$$

it follows that $\zeta_m(s)$ is holomorphic at $s = 1$, since the only possible singularity in the product is from $L(s, 1)$, and this is cancelled out by the zero of $L(1, \chi)$.

Hence ζ_m would be holomorphic at $s = 1$, and by Propositions 3.4 and 3.5 (already established), it would be holomorphic in the entire half-plane $\Re(s) > 0$.

However, since $\zeta_m(s)$ is a Dirichlet series with positive coefficients, by Proposition 2.8 this would imply that the series converges for all $\Re(s) > 0$. Let us consider the Euler product for $\zeta_m(s)$. The p -th Euler factor is

$$\frac{1}{(1 - p^{-f(p)s})^{g(p)}} = \left(1 + p^{-f(p)s} + p^{-2f(p)s} + \dots \right)^{g(p)},$$

which dominates the series

$$1 + p^{-\varphi(m)s} + p^{-2\varphi(m)s} + \dots.$$

This is because $\varphi(m) > f(p)$, hence $p^{-\varphi(m)s} < p^{-f(p)s}$, and so term-by-term we have:

$$1 + p^{-\varphi(m)s} + p^{-2\varphi(m)s} + \dots \leq 1 + p^{-f(p)s} + p^{-2f(p)s} + \dots \leq (1 + p^{-f(p)s} + p^{-2f(p)s} + \dots)^{g(p)}.$$

Thus, we obtain:

$$\prod_{p \nmid m} (1 + p^{-\varphi(m)s} + p^{-2\varphi(m)s} + \dots) \leq \prod_{p \nmid m} \left(\frac{1}{1 - p^{-f(p)s}} \right)^{g(p)} = \zeta_m(s).$$

But the left-hand side is equal to

$$\sum_{\substack{n \geq 1 \\ (n, m) = 1}} \frac{1}{n^{\varphi(m)s}}.$$

This is a Dirichlet series which dominates a multiple of the classical series $\sum_{n=1}^{\infty} \frac{1}{n^{\varphi(m)s}}$, restricted to $(n, m) = 1$.

Now set $s = \frac{1}{\varphi(m)} > 0$. Then we get:

$$\sum_{\substack{n \geq 1 \\ (n, m) = 1}} \frac{1}{n} \leq \zeta_m \left(\frac{1}{\varphi(m)} \right).$$

But the series $\sum_{(n, m) = 1} \frac{1}{n}$ diverges, since it dominates $\sum_{p \nmid m} \frac{1}{p}$, and the sum over primes diverges.

Therefore, $\zeta_m(s)$ must diverge at $s = \frac{1}{\varphi(m)}$, contradicting the assumption that it is convergent for all $\Re(s) > 0$.

This contradiction implies our assumption was false. Hence, for all nontrivial characters $\chi \neq 1$, we must have

$$L(1, \chi) \neq 0.$$

□

4 Density and Dirichlet Theorem

Let $\mathcal{P}$ be a set of prime numbers. When $s \rightarrow 1$, with $s \in \mathbb{R}$ (to fix the idea), we have from earlier observation

$$\sum_{p \in \mathcal{P}} \frac{1}{p^s} \sim \log \left(\frac{1}{s-1} \right) \quad \text{as } s \rightarrow 1^+,$$

Density

Let $A \subseteq \mathcal{P}$ be a subset of prime numbers. One says that A has density equal to a real number $k \in [0, 1]$ if

$$\lim_{s \rightarrow 1^+} \frac{\sum_{p \in A} \frac{1}{p^s}}{\log \left(\frac{1}{s-1} \right)} = k.$$

Thus, the density of any subset $A \subseteq \mathcal{P}$ always lies in the interval $[0, 1]$.

Let χ be a character of $G(m)$. Define

$$f_\chi(s) = \sum_{p \nmid m} \frac{\chi(\bar{p})}{p^s}.$$

This Dirichlet series converges for $\Re(s) > 1$.

Lemma 4.1. *If $\chi = \chi_0 = 1$ is the trivial character, then*

$$f_\chi(s) \sim \log\left(\frac{1}{s-1}\right) \quad \text{as } s \rightarrow 1^+.$$

Proof. By Corollary 3.2 we know that

$$\sum_{p \in \mathcal{P}} \frac{1}{p^s} \sim \log\left(\frac{1}{s-1}\right) \quad \text{as } s \rightarrow 1^+. \quad (1)$$

We also write

$$\begin{aligned} \sum_{p \in \mathcal{P}} \frac{1}{p^s} &= \sum_{\substack{p \in \mathcal{P} \\ p \nmid m}} \frac{1}{p^s} + \sum_{\substack{p \in \mathcal{P} \\ p \mid m}} \frac{1}{p^s} \\ &= \sum_{p \nmid m} \frac{1(\bar{p})}{p^s} + \sum_{p \mid m} \frac{1(\bar{p})}{p^s}. \end{aligned}$$

Let $A := \{p \in \mathcal{P} : p \mid m\}$. This is a finite set of primes.

Taking the limit in (1), we have:

$$\lim_{s \rightarrow 1^+} \frac{\sum_{p \in \mathcal{P}} \frac{1}{p^s}}{\log\left(\frac{1}{s-1}\right)} = \lim_{s \rightarrow 1^+} \left(\frac{\sum_{p \nmid m} \frac{1(\bar{p})}{p^s}}{\log\left(\frac{1}{s-1}\right)} + \frac{\sum_{p \in A} \frac{1}{p^s}}{\log\left(\frac{1}{s-1}\right)} \right).$$

Since A is finite, the second term tends to zero as $s \rightarrow 1^+$. Thus,

$$1 = \lim_{s \rightarrow 1^+} \frac{f_\chi(s)}{\log\left(\frac{1}{s-1}\right)} + 0,$$

which implies that

$$f_\chi(s) \sim \log\left(\frac{1}{s-1}\right) \quad \text{as } s \rightarrow 1^+.$$

□

Lemma 4.2. *If $\chi \neq 1$, then $f_\chi(s)$ remains bounded as $s \rightarrow 1$.*

Proof. We have seen that For $\chi \neq 1$, the series $L(s, \chi)$ converges (in fact, converges absolutely) in the half-plane $\Re(s) > 0$. In particular, for $\Re(s) > 1$, we have

$$L(s, \chi) = \prod_{p \in \mathbb{P}} \frac{1}{1 - \frac{\chi(p)}{p^s}}.$$

Taking logarithms, we obtain for $\Re(s) > 1$,

$$\log L(s, \chi) = \sum_{p \in \mathbb{P}} \log\left(\frac{1}{1 - \frac{\chi(p)}{p^s}}\right) = - \sum_{p \in \mathbb{P}} \log\left(1 - \frac{\chi(p)}{p^s}\right).$$

Using the identity $\log(1 - x) = - \sum_{n=1}^{\infty} \frac{x^n}{n}$ (valid for $|x| < 1$), we get

$$\log L(s, \chi) = \sum_{p \in \mathbb{P}} \sum_{n=1}^{\infty} \frac{(\chi(p))^n}{np^{ns}} = \sum_{n=1}^{\infty} \sum_{p \in \mathbb{P}} \frac{(\chi(p))^n}{np^{ns}}.$$

Split the sum into two parts:

$$\log L(s, \chi) = \sum_{p \in \mathbb{P}} \frac{\chi(p)}{p^s} + \sum_{n \geq 2} \sum_{p \in \mathbb{P}} \frac{(\chi(p))^n}{np^{ns}} = f_\chi(s) + F_\chi(s),$$

where we define

$$F_\chi(s) := \sum_{n \geq 2} \sum_{p \in \mathbb{P}} \frac{(\chi(p))^n}{np^{ns}}.$$

Claim 1: $\log L(s, \chi)$ is bounded as $s \rightarrow 1$.

Claim 2: $F_\chi(s)$ is bounded as $s \rightarrow 1$.

Hence, from Theorem 3.9 and Proposition 3.6, we conclude that if $\chi \neq 1$, then $L(s, \chi)$ is bounded and nonzero as $s \rightarrow 1$, and thus

$$\log L(s, \chi) \text{ is bounded as } s \rightarrow 1.$$

So it suffices to prove Claim 2.

Proof of Claim 2. Consider

$$F_\chi(s) = \sum_{n \geq 2} \sum_{p \in \mathbb{P}} \frac{(\chi(p))^n}{np^{ns}}.$$

Since χ is a Dirichlet character, each value $\chi(p)$ lies in the unit circle, i.e., $|\chi(p)| = 1$. Therefore,

$$\left| \frac{(\chi(p))^n}{np^{ns}} \right| = \frac{1}{np^{n\Re(s)}}.$$

Assuming s is real and $s > 1$ (we can restrict to this case), we estimate

$$|F_\chi(s)| \leq \sum_{n \geq 2} \sum_{p \in \mathbb{P}} \frac{1}{np^{ns}}.$$

Now, by Corollary 3.2, the double sum

$$\sum_{\substack{p \in \mathbb{P} \\ n \geq 2}} \frac{1}{np^{ns}}$$

is bounded as $s \rightarrow 1$. Indeed, note that

$$\frac{1}{np^{ns}} \leq \frac{1}{p^{ns}} \quad \text{for } n \geq 2,$$

so

$$|F_\chi(s)| \leq \sum_{n \geq 2} \sum_{p \in \mathbb{P}} \frac{1}{np^{ns}} \leq \sum_{p \in \mathbb{P}} \sum_{n \geq 2} \frac{1}{p^{ns}}.$$

Since

$$\sum_{n \geq 2} \frac{1}{p^{ns}} = \frac{1}{p^{2s}} + \frac{1}{p^{3s}} + \cdots = \frac{1}{p^{2s}(1 - 1/p^s)} \leq \frac{C}{p^{2s}} \quad (\text{for some constant } C \text{ as } s \rightarrow 1),$$

we have

$$\sum_{p \in \mathbb{P}} \sum_{n \geq 2} \frac{1}{p^{ns}} \leq \sum_{p \in \mathbb{P}} \frac{C}{p^{2s}},$$

which converges and is bounded for $s > 1$ near 1.

Hence $F_\chi(s)$ is bounded as $s \rightarrow 1$. Therefore,

$$f_\chi(s) = \log L(s, \chi) - F_\chi(s)$$

is also bounded as $s \rightarrow 1$.

This completes the proof of the lemma. □

We now study the behavior of the function

$$g_a(s) := \sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{m}}} \frac{1}{p^s}, \quad \text{as } s \rightarrow 1^+.$$

Lemma 4.3. *One has*

$$g_a(s) = \frac{1}{\varphi(m)} \sum_{\chi \in \widehat{G(m)}} \chi(a^{-1}) f_\chi(s),$$

where the sum is extended over all Dirichlet characters χ modulo m .

Proof. We begin with the right-hand side:

$$\frac{1}{\varphi(m)} \sum_{\chi \in \widehat{G(m)}} \chi(a^{-1}) f_\chi(s) = \frac{1}{\varphi(m)} \sum_{\chi \in \widehat{G(m)}} \chi(a^{-1}) \sum_{p \in \mathbb{P}, p \bmod m} \frac{\chi(p)}{p^s}.$$

Rewriting the order of summation, we obtain:

$$g_a(s) = \frac{1}{\varphi(m)} \sum_{p \in \mathbb{P}, p \bmod m} \left(\sum_{\chi \in \widehat{G(m)}} \chi(a^{-1}) \chi(p) \right) \frac{1}{p^s}.$$

By the orthogonality relations for characters, for any $n \in G(m)$,

$$\sum_{\chi \in \widehat{G(m)}} \chi(n) = \begin{cases} \varphi(m), & \text{if } n \equiv 1 \pmod{m}, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$\sum_{\chi \in \widehat{G(m)}} \chi(a^{-1}p) = \begin{cases} \varphi(m), & \text{if } p \equiv a \pmod{m}, \\ 0, & \text{otherwise.} \end{cases}$$

Hence, the sum simplifies to:

$$g_a(s) = \sum_{\substack{p \in \mathbb{P} \\ p \equiv a \pmod{m}}} \frac{1}{p^s}.$$

Thus,

$$g_a(s) = \frac{1}{\varphi(m)} \sum_{\chi \in \widehat{G(m)}} \chi(a^{-1}) f_\chi(s),$$

as claimed. The sum is extended over all Dirichlet characters χ modulo m . □

Theorem 4.4 (Dirichlet's Theorem on Primes in Arithmetic Progressions). *Let $m \geq 1$, and let $a \in \mathbb{Z}$ be such that $\gcd(a, m) = 1$. Let*

$$\mathcal{P}_a = \{p \in \mathcal{P} : p \equiv a \pmod{m}\}.$$

Then the set $\mathcal{P}_a$ has density

$$\frac{1}{\varphi(m)}.$$

In other words, the prime numbers are “equally distributed” among the residue classes modulo m that are coprime to m .

Proof. By Lemma 4.3, we have

$$\sum_{p \in \mathcal{P}_a} \frac{1}{p^s} = \frac{1}{\varphi(m)} \sum_{\chi \in \widehat{G(m)}} \chi(a^{-1}) f_\chi(s).$$

Multiplying both sides by $\varphi(m)$, we obtain

$$\varphi(m) \sum_{p \in \mathcal{P}_a} \frac{1}{p^s} = \sum_{\substack{\chi \in \widehat{G(m)} \\ \chi=1}} \chi(a^{-1}) f_\chi(s) + \sum_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} \chi(a^{-1}) f_\chi(s).$$

Since $\chi = 1$ is the principal character and $\chi(1) = 1$, this becomes

$$\varphi(m) \sum_{p \in \mathcal{P}_a} \frac{1}{p^s} = f_{\chi=1}(s) + \sum_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} \chi(a^{-1}) f_\chi(s).$$

Dividing both sides by $\log \frac{1}{s-1}$, we have

$$\frac{\varphi(m)}{\log \frac{1}{s-1}} \sum_{p \in \mathcal{P}_a} \frac{1}{p^s} = \frac{f_{\chi=1}(s)}{\log \frac{1}{s-1}} + \frac{1}{\log \frac{1}{s-1}} \sum_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} \chi(a^{-1}) f_\chi(s).$$

By Lemma 4.1, we know that $f_{\chi=1}(s) \sim \log \frac{1}{s-1}$ as $s \rightarrow 1$, so

$$\frac{f_{\chi=1}(s)}{\log \frac{1}{s-1}} \rightarrow 1 \quad \text{as } s \rightarrow 1.$$

By Lemma 4.2, for $\chi \neq 1$, the functions $f_\chi(s)$ remain bounded as $s \rightarrow 1$, and since $|\widehat{G(m)}| = \varphi(m)$ is finite, the sum

$$\sum_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} \chi(a^{-1}) f_\chi(s)$$

is a finite sum of bounded terms and thus remains bounded as $s \rightarrow 1$. Hence,

$$\left| \sum_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} \chi(a^{-1}) f_\chi(s) \right| \leq \sum_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} |f_\chi(s)| < \varphi(m) \cdot \alpha,$$

where

$$\alpha := \max_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} \sup_{s \in (1,2)} |f_\chi(s)|.$$

Thus,

$$\frac{1}{\log \frac{1}{s-1}} \sum_{\substack{\chi \in \widehat{G(m)} \\ \chi \neq 1}} \chi(a^{-1}) f_\chi(s) \rightarrow 0 \quad \text{as } s \rightarrow 1.$$

Combining the above, we conclude:

$$\frac{\varphi(m)}{\log \frac{1}{s-1}} \sum_{p \in P_a} \frac{1}{p^s} \rightarrow 1 \quad \text{as } s \rightarrow 1,$$

or equivalently,

$$\sum_{p \in P_a} \frac{1}{p^s} \sim \frac{1}{\varphi(m)} \log \frac{1}{s-1} \quad \text{as } s \rightarrow 1,$$

which implies that the set P_a has density $\frac{1}{\varphi(m)}$. □

Corollary 4.1. *The set P_a is infinite.*

Proof. Suppose, for contradiction, that the set P_a is finite. Then the series

$$\sum_{p \in P_a} \frac{1}{p^s}$$

is finite for all $s > 1$. Hence, we would have

$$\lim_{s \rightarrow 1^+} \frac{1}{\log \frac{1}{s-1}} \sum_{p \in P_a} \frac{1}{p^s} = 0,$$

which implies that the density of P_a is zero.

However, by Theorem 3.12, we have already shown that

$$\sum_{p \in P_a} \frac{1}{p^s} \sim \frac{1}{\varphi(m)} \log \frac{1}{s-1} \quad \text{as } s \rightarrow 1,$$

so that

$$\lim_{s \rightarrow 1^+} \frac{1}{\log \frac{1}{s-1}} \sum_{p \in P_a} \frac{1}{p^s} = \frac{1}{\varphi(m)} > 0.$$

This contradiction implies that the assumption that P_a is finite must be false. Therefore, P_a is infinite. □

References

- [1] Jean-Pierre Serre, *A Course in Arithmetic*, Springer-Verlag, Graduate Texts in Mathematics, vol. 7, 1996.
- [2] Elias M. Stein and Rami Shakarchi, *Complex Analysis*, Princeton University Press, Princeton Lectures in Analysis, vol. 2, 2010.